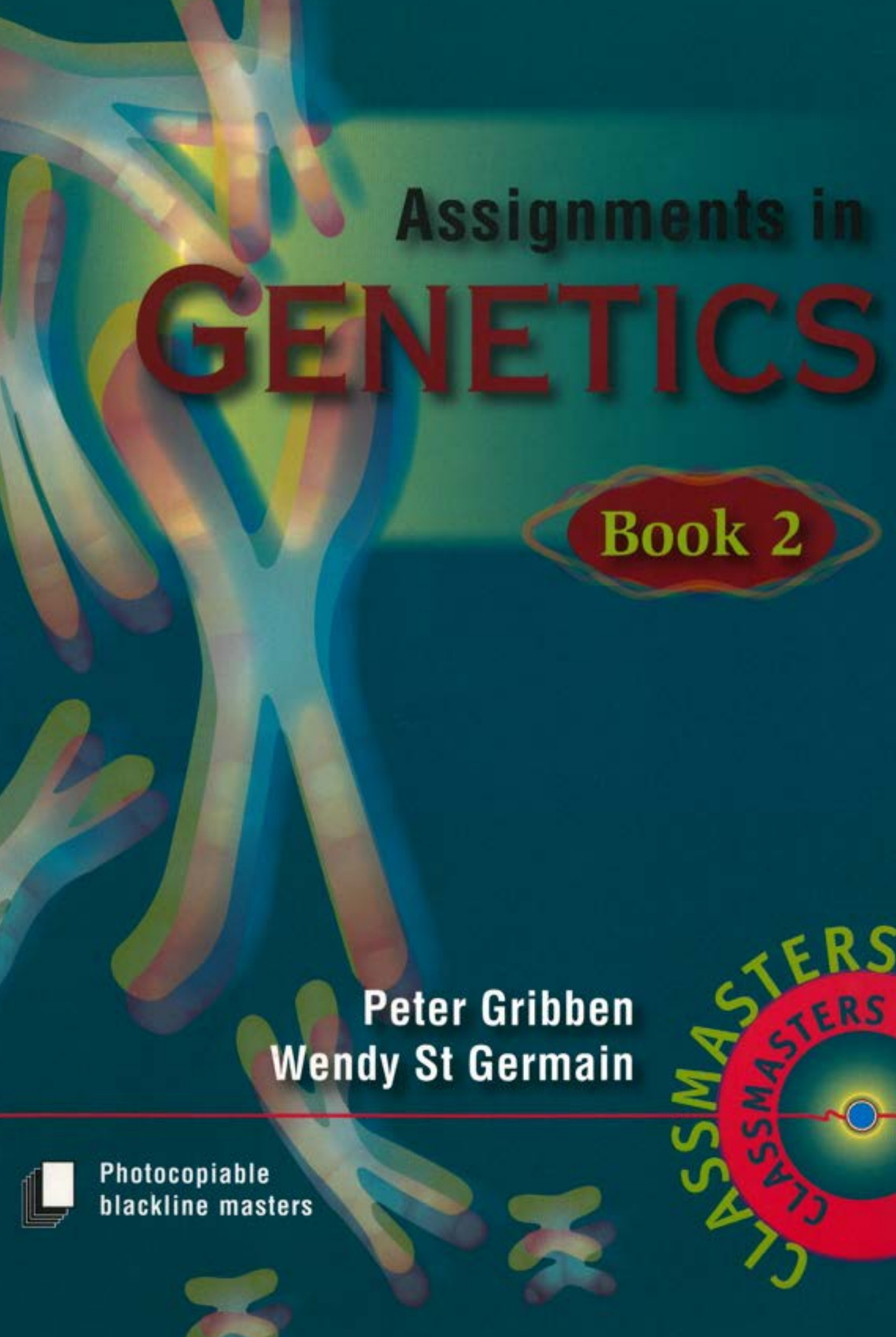




Assignments in **GENETICS**

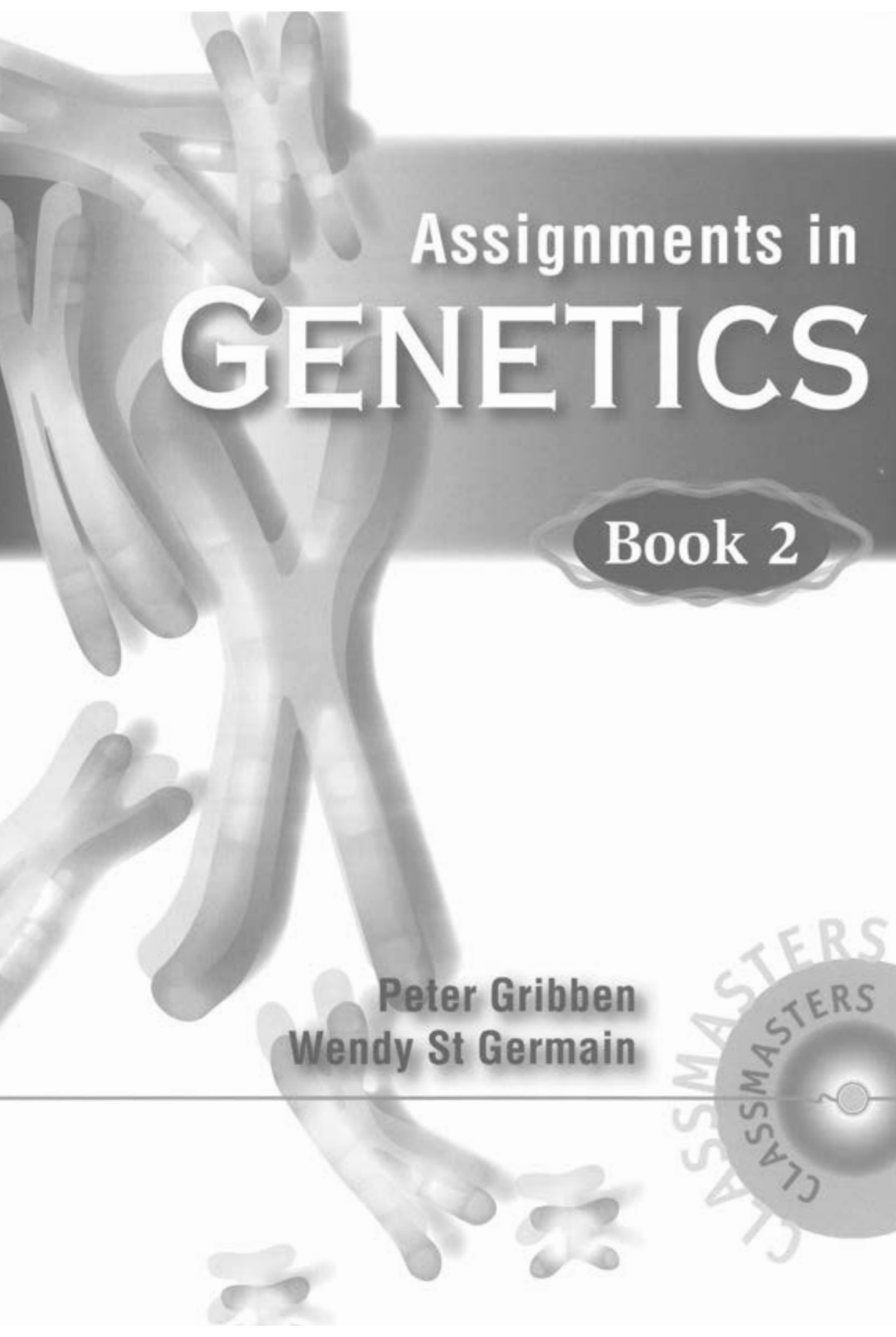
Book 2

**Peter Gribben
Wendy St Germain**



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Assignments in **GENETICS**

Book 2

**Peter Gribben
Wendy St Germain**



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Introduction

Genetics is a topic that is increasingly of interest to teachers, students and the general public. It has become an important part of high school science syllabuses, both in junior and senior courses. While *Instant Lessons in Genetics book 2* has been written with the needs of senior students in mind, teachers of junior classes will find the work sheets useful as extension activities for their more capable students.

Instant Lessons in Genetics book 2 has been specifically written so that the extensive technical terms and concepts used in Genetics are introduced through a variety of activities, making it easier for students to understand and remember each concept. In keeping with this, a considerable portion of this book contains 'stand alone' exercises, requiring little or no previous knowledge of the subject. Other sections of this book can be used to assess each student's progress and knowledge of Genetics, and to identify areas of weakness. Selected answers have been provided for each work sheet.

Instant Lessons in Genetics book 2 builds student knowledge, confidence and interest in a fascinating, if occasionally challenging subject.



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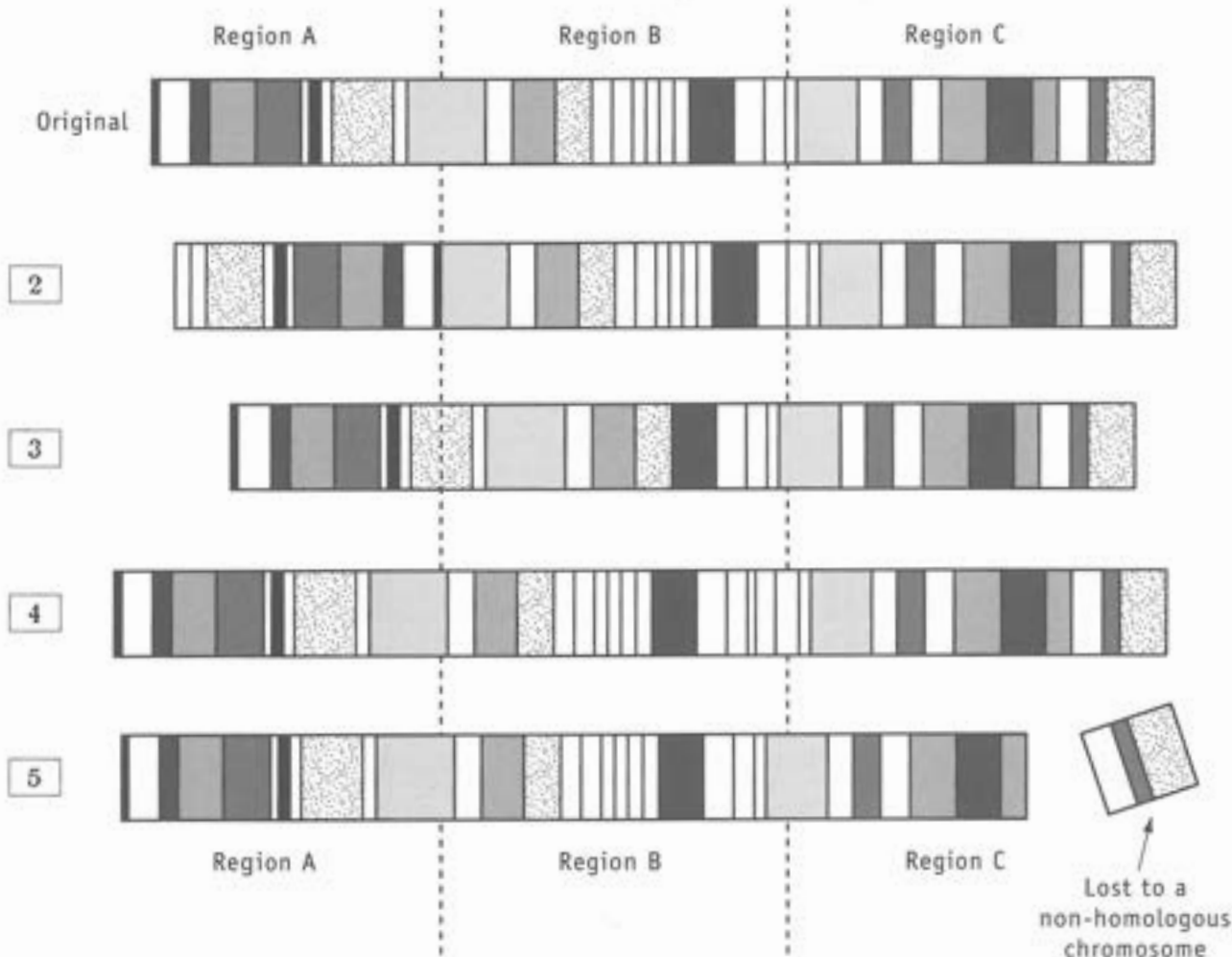
Throughout evolution, random changes have occurred to DNA. These changes can be harmful or beneficial, depending on how, and in what sort of environment, they are expressed. For example consider the following scenario. A species of bird has toes. Among this population, through mutation, some birds carry a gene producing webbed feet. Possession of webbed feet would improve swimming ability and ability to walk through marshlands. This would enable these birds to be more successful feeders in wet areas. The webbed-feet birds would breed successfully in wetlands and the trait would increase through time.

1 In what ways might the possession of webbed feet be a disadvantage for these birds?

Alternatively, a mutation might debilitate (weaken) an organism. For example, the ability to produce a vital digestive organism may be reduced or prevented. This would be an example of a harmful mutation. There are four main mechanisms involved in mutation:

- *deletion*, the removal and loss of genetic material
- *addition/duplication*, adding extra genetic material onto a chromosome. When this occurs between homologous pairs (the most common occurrence) it causes the duplication of genes already present on the chromosome.
- *translocation*, the movement of genetic material from one chromosome onto another non-homologous chromosome
- *inversion*, inverting information on a gene. This occurs when a chromosome breaks and reforms, but with the gene sequence inverted.

Compare the strands of DNA in examples 2–5 below with the original provided. Name the type of mutation that has occurred each time. Indicate the region (A, B or C) in which the mutation occurred.





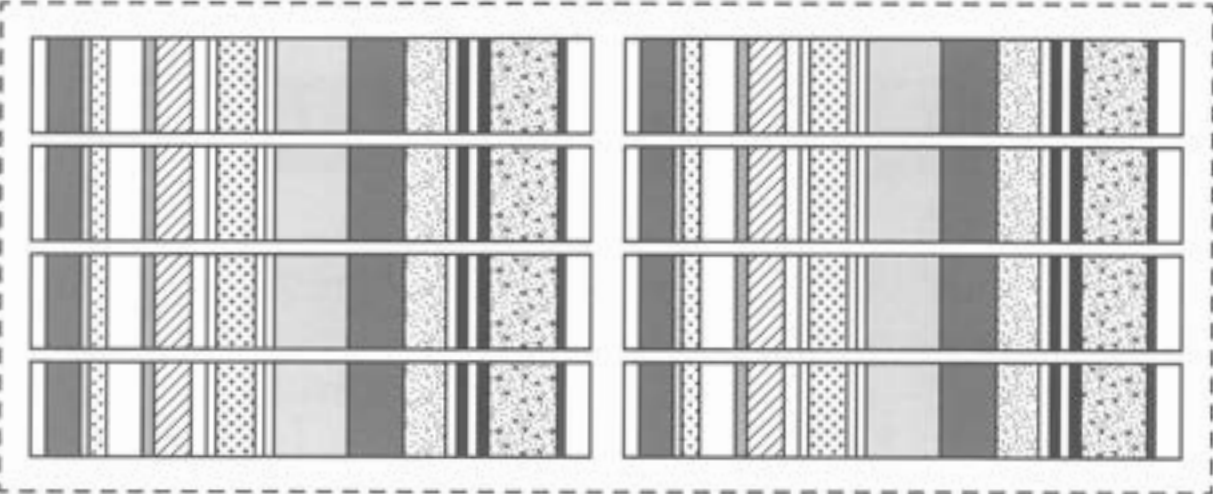
Recognising Mutations

33

It is recommended that you complete worksheet 32, 'More About Mutations: changing DNA' before you attempt this sheet.

Below are eight identical strands of DNA. Colour them, using several colours to highlight the different gene sequences. Each strand should be coloured in the same way, they should all look identical. Cut the strands out. Glue one of the strands in each of the spaces labelled 'Before mutation'. Then make the necessary changes to each of the remaining strips, to illustrate the mutations required. When you have made deletions, make sure you also glue the deleted piece in the box, away from the chromosome.

Addition / duplication	Before mutation:	
	After mutation:	
Deletion	Before mutation:	
	After mutation:	
Inversion	Before mutation:	
	After mutation:	
Translocation	Before mutation:	
	After mutation:	





Select the correct word from the pair provided each time, to make the following passage read properly.

In the past, it was recognised that there were often similarities among relatives. These resemblances couldn't be explained until a monk named Gregor (1: Mendel/Mendeleyev) experimented with generation after generation of (2: peas/pears). Each generation had recognisable features to help calculate the expected (3: frequent/frequencies) of the characteristics he studied. His contemporary, (4: naturalist/naturopath) Charles Darwin, produced a paper describing his theory about *how* similarities within families occurred.

The transmission of specific characteristics, or traits, from parents to offspring is known as (5: heredity/genealogy). At the same time as Darwin presented his 'Theory of Natural (6: Selection/Inheritance)', Alfred Wallace submitted an almost identical paper.

During sexual reproduction, traits are passed from parent to offspring, with each parent contributing (7: half/all) of their genetic material to their offspring. This genetic information is stored in the (8: cytoplasm/nucleus) of a cell in the form of a (9: simple/complex) compound called DNA. Traits can be dominant or recessive. In the case of (10: recessive/dominant) traits only a 'single dose' is needed for a characteristic to be *expressed* in an individual's phenotype, that is its (11: visible/genetic) properties. If the trait is (12: recessive/dominant) a 'double dose' is required for it to be expressed. Recessive traits are usually only expressed in the (13: absence/presence) of dominant traits.

It is important to understand the difference between phenotype and genotype when studying genetics. (14: Genotype/Phenotype) is the term used to describe *observable* features, such as hair colour and behaviour. The genetic make-up of an individual, whether it is expressed or not, is called its (15: genotype/phenotype).

The study of genetics can be divided into three main branches: Mendelian genetics, molecular genetics and population genetics. The study of what the gene is made of, how it is controlled and what it produces as a result of how it is put together is known as (16: population/molecular) genetics. (17: Mendelian/Population) genetics studies the genetics of whole groups rather than the genetics of individuals. (18: Mendelian/Molecular) genetics is also known as classical genetics.

Did you know?: Darwin wanted Wallace also to be credited for the theory of selection but, recognising the years Darwin spent in collecting the necessary evidence, Wallace refused the offer.



Charles Darwin



Reciprocal cross

Whenever we cross a male possessing a specific trait with a female possessing a different trait, the procedure is called a *reciprocal cross*. Mendel created reciprocal crosses when he crossed female yellow peas with male green peas.

- 1 What would have been an alternative reciprocal cross using the same characteristics?
-

Filial: to do with a son, daughter, child or offspring.

F₁ and F₂ generations

This is a shorthand term for 'first filial' and 'second filial' generations. The first generation progeny (offspring) of a reciprocal cross are called the F₁ generation. The abbreviation F₁ means, '*The resulting progeny of the reciprocal cross being studied*'.

- 2 Why do we usually say F₁ rather than use the complete definition given above?
-

The F₂ generation is the progeny of crossing the F₁ individuals. Members of the F₁ generation can be allowed to self-fertilise (in plants) or cross-fertilise with other members of the F₁ generation. Either way, the progeny is referred to as the F₂ generation.

Cell cycle: growth and reproduction

All cells grow and reproduce. They use either of the processes mitosis or meiosis.

Mitosis occurs in *somatic* cells. These are cells used for growth and development of an organism.

Meiosis is exclusively used by *germ* cells. Germ cells are involved in reproduction. Male and female reproductive cells are referred to as *gametes*. When a female gamete (ovum) is fertilised by a male gamete (sperm) a *zygote* is formed.

Use terms mentioned previously on this sheet and match them to the following definitions/descriptions.

- 3 Takes place in the nucleus of a dividing cell and forms two new nuclei that have the same number of chromosomes as the parent nucleus. _____
- 4 A process causing the number of chromosomes in gamete-producing cells to be halved. _____
- 5 A mature male or female germ cell. _____
- 6 A type of cell that makes up tissues and organs in the body, apart from germ cells. _____
- 7 Results from the fusion of a male sex cell and a female sex cell. _____
- 8 Male gamete or germ cell. _____
- 9 Female gamete or germ cell. _____
- 10 Offspring of plants or animals. _____
- 11 Paired crosses where a male with a specific trait is crossed with a female possessing another trait, and vice versa. _____

Mitosis: occurs in somatic cells and produces identical cells. Exact chromosome duplication occurs, and two daughter cells are formed. Mitosis has four phases: prophase, metaphase, anaphase and telophase (occurring in that order).

Cut out the box above and glue it in your book as your title and introduction. Cut out the diagrams (provided on sheet 36B) and the descriptions below of the phases of mitosis. Match the diagrams with the correct descriptions and stick them beside each other in your book. The summary goes at the end. (Hint: the summary will help you arrange and match the boxes).

When all chromatids are aligned at the 'equator', they begin to pull apart. The 'twins' will now go their separate ways, half to the north 'pole' and half to the south. The centromere goes first, as if it were being pulled by a magnet. The 'arms' of the chromatid follow.

The longest phase. At the beginning of mitosis, chromosomes become more compact (condense) and undergo replication. When they decondense, we find two replicas of what was originally one chromosome. These 'twins' are now referred to as *chromatids*. Chromatids are joined by a link called a *centromere* (sometimes called a 'kinetochore' in older text books.) The centromere 'pinches' the twins together, forming the 'arms' on the chromatids. The short arm is called the '*p*-arm' and the long arm is called the '*q*-arm'.

The final phase. Genetic material has migrated to the poles. A new nuclear envelope reforms around each daughter cell. During this phase, the cell begins to pinch itself at the 'equator' to form two smaller, new cells. This process is called *cleavage*. (From the word 'cleave', meaning 'to cut'.) Half of the original cytoplasm goes into each cell. It is important to include some of the original cytoplasm in the new cells because it contains many organelles responsible for ensuring the genetic material within the nucleus can be used properly. The genetic material (DNA) can be compared to an architect's blueprint, it must be followed exactly.

Spindles form. Spindles are made of protein fibres and microtubules. As the name suggests, microtubules are tiny, tube-like structures. The spindles form an oval structure, along which chromatids can align themselves as they migrate to the 'equator'.

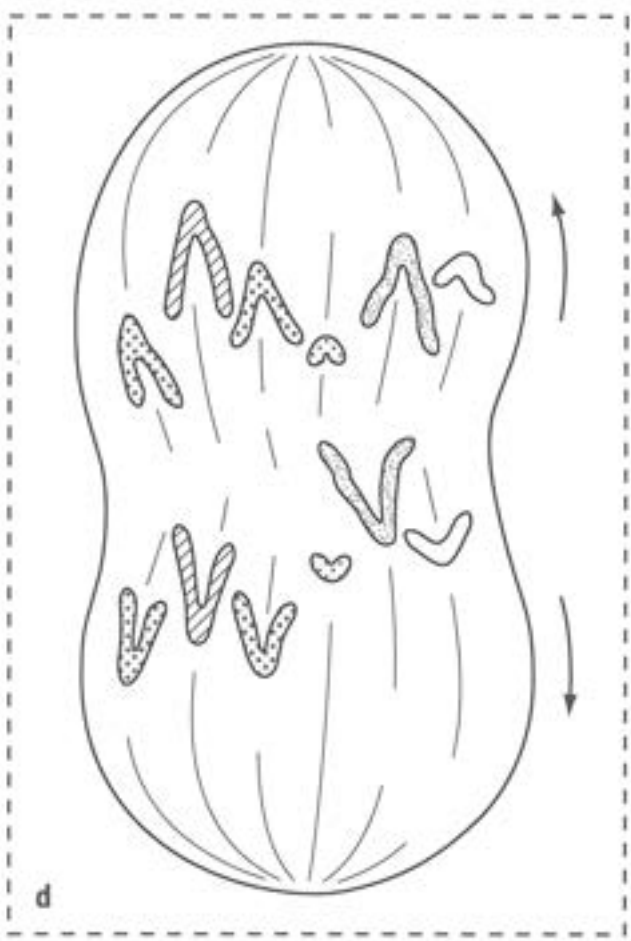
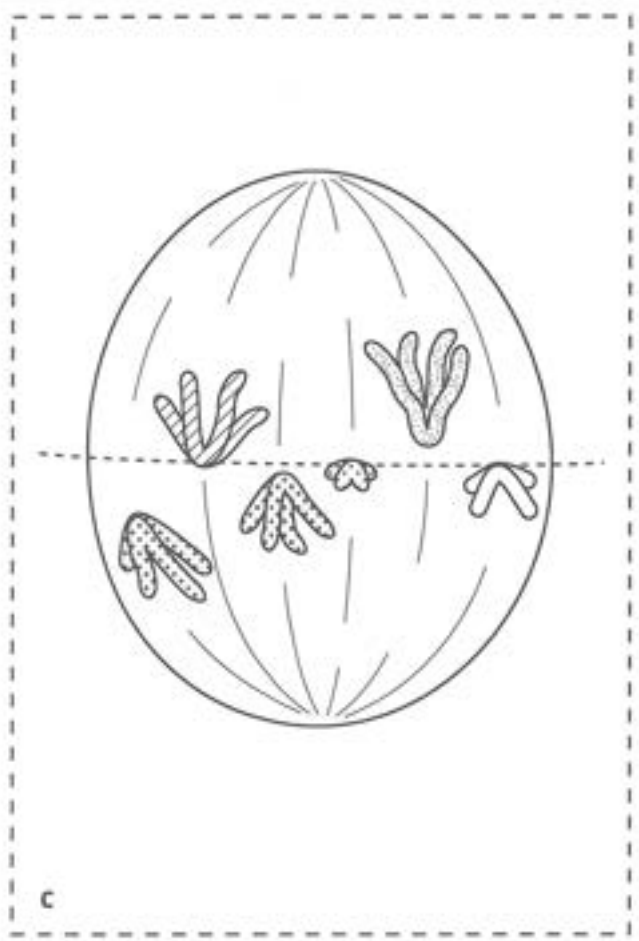
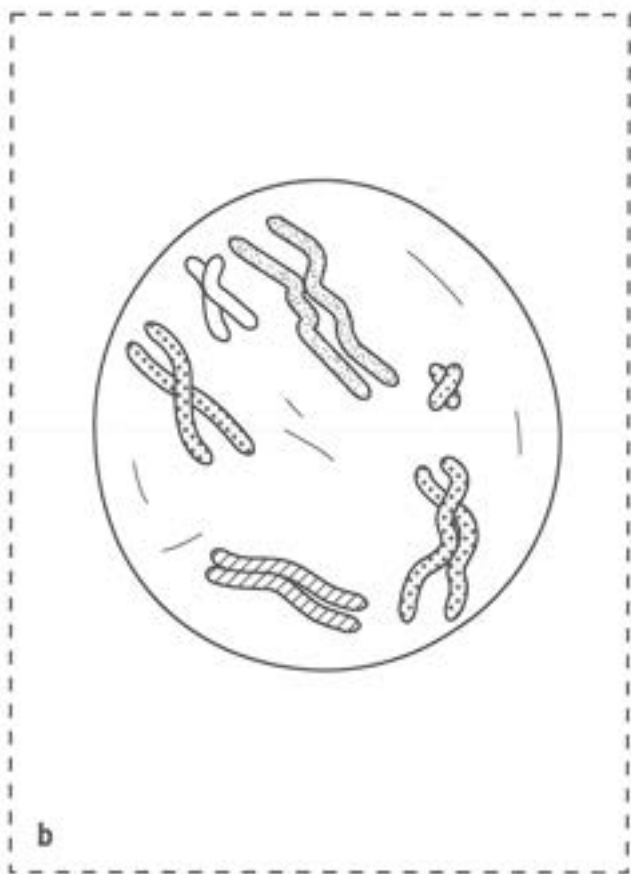
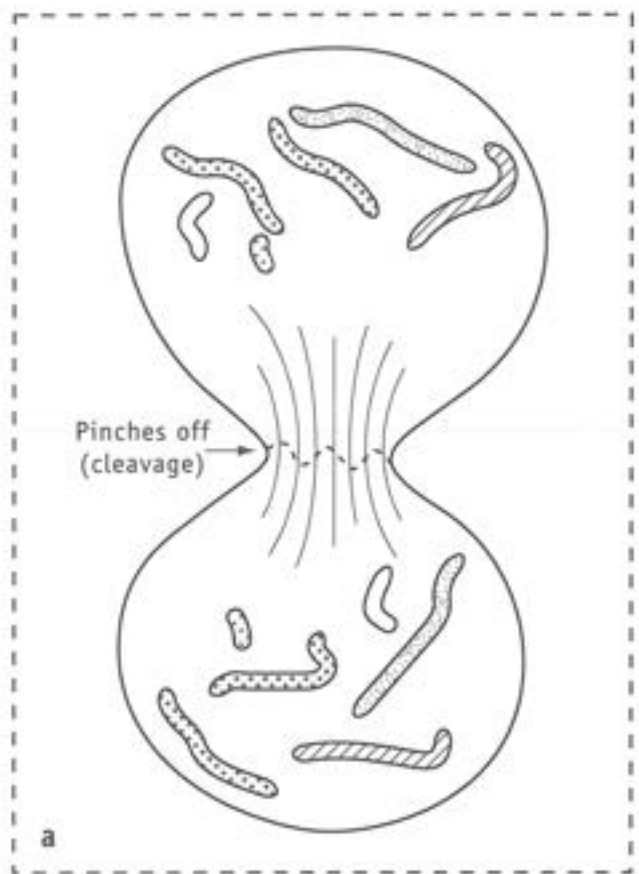
Summary

Prophase: the earliest stage of nuclear division. Chromosomes condense, replicate and decondense.

Metaphase: the chromosomes start aligning themselves along the 'equator' of the nucleus.

Anaphase: when chromosomes are aligned at the 'equator', they then move towards the poles.

Telophase: the nuclear membrane reforms, enclosing the chromosomes in newly formed daughter cells.



Meiosis (from the Greek word, meaning 'to make smaller') is a more complicated procedure than mitosis. Both processes involve four phases: prophase, metaphase, anaphase and telophase. Meiosis undergoes the process twice and, for this reason, the phases are labelled: prophase I, metaphase I, etc. and prophase II, metaphase II, etc. to signify the stage at which the cell is at a given time.

Meiosis divides the nucleus and its contents once, then repeats the procedure a second time. By dividing the nuclear material twice, the original diploid cell is divided and reassembled as four haploid cells. Because of this division, and the 'crossing-over' of genetic material between chromatids, the four daughter cells are all genetically different.

In order to help geneticists identify them, chromosomes are numbered in every species. Humans have 46 chromosomes (23 pairs). This is written as $2n = 46$, because both parents give us 23 chromosomes, which totals 46. The numbered chromosomes form homologues.

For example: chromosome 1 will be a certain length and will contain specific genes, perhaps including one for eye colour. Its homologue, also called chromosome 1, will be the same length and contain the eye colour gene (but not necessarily coding for the same eye colour). One of the chromosomes will have been provided by the mother, the other from the father. For each of the 23 chromosomes inherited from one parent there is (or should be) a homologue. During meiosis, homologous pairs come together, including a pair of sex chromosomes.

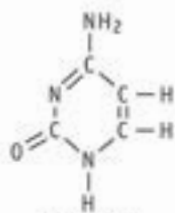
Because humans have 23 pairs of chromosomes, the random way in which chromosomes are divided in meiosis could theoretically give up to 2^{23} different combinations for gametes. The 23rd pair in each set of human chromosomes is assigned by geneticists to the sex chromosomes, XX for females and XY for males.

The number of chromosomes doesn't always match up with how developed an organism is. A platypus has 52 chromosomes and a domestic dog has 78. Both have more chromosomes than humans.

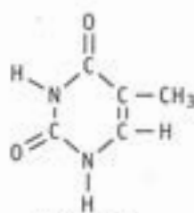
- 1 Describe two similarities between mitosis and meiosis.
- 2 Describe two differences between mitosis and meiosis.
- 3 Rearrange the phases of meiosis in the correct order, from first to last:
anaphase, telophase, prophase, metaphase.
- 4 Which type of cell division maintains the diploid number?
- 5 Why are chromosomes numbered?
- 6 'Have the same genes (or alleles) with genetic loci arranged in the same order.' What does this describe?
- 7 'Mitosis involves more steps than meiosis.' True or false?
- 8 'The cells produced by meiosis are genetically identical.' True or false?
- 9 'Meiosis could produce more than 8 000 000 000 possible gametes.' True or false?
- 10 'Diploid cells have twice the number of chromosomes of haploid cells.' True or false?
- 11 'All mammals have a diploid number of 46.' True or false?
- 12 'Similar chromosomes are called homologous chromosomes.' True or false?
- 13 'Possible combinations for sex chromosomes include XX, XY and YY.' True or false?
- 14 'In general, the more complex an organism, the greater the number of chromosomes it has.' True or false?



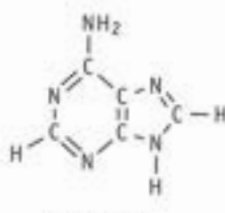
DNA can do many things, so it is amazing that it is made up of only four nitrogen-based substances (bases), sugar and phosphate molecules. The four nitrogen bases are cytosine, thymine, adenine and guanine. Cytosine and thymine are *pyrimidines*, and adenine and guanine are *purines*. Pyrimidines and purines have specific shapes.



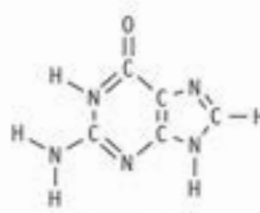
Cytosine



Thymine



Adenine

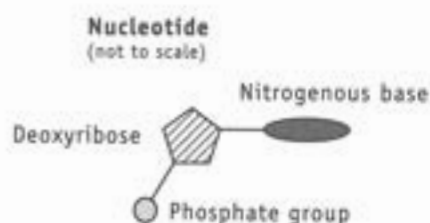


Guanine

1 What elements are present in the compounds shown above?

2 What is a major difference between purines and pyrimidines?

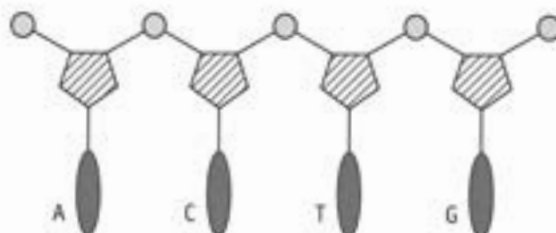
The nitrogen base, sugar and phosphate form a three-part unit called a *nucleotide*. Each base is attached to one sugar and one phosphate molecule. The sugar involved is *deoxyribose*, hence the name *deoxyribonucleic acid* or DNA. (Deoxyribose means that the sugar is ribose, but has an oxygen atom missing). The diagram on the right shows the general structure of a nucleotide.



Study the portion of a single chain of DNA shown below. Nucleotides are held together by bonds between the sugar and phosphate groups. The nitrogen bases (adenine [A], cytosine [C], guanine [G] and thymine [T]) are side groups.

3 Copy the chain of DNA shown. (Leave plenty of space around it.)

4 Label a phosphate and a deoxyribose group on your copy of the DNA chain.



This chain will be loosely linked by hydrogen bonds to another (parallel) chain. Cytosine links with guanine, and adenine links with thymine.

5 Draw another chain of DNA parallel to and linked with the one you drew for Question 3. Identify the base used in each nucleotide.

When DNA is analysed it is found that the amounts of adenine and thymine are always equal to each other, and that the amounts of cytosine and guanine are equal to each other.

6 Explain why the amounts discussed above are equal.

The sequence of amino acids can change, by mutation for example. Mistakes can occur spontaneously — some estimates say in about one in every 10^9 replications of base pairs. Some factors can increase the rate of mutations. High energy radiation, such as ultraviolet radiation and X-rays, can cause mutations. Some chemicals have also been suspected of causing mutations. These are called *mutagens*.

7 Write 10^9 as a number in expanded form and in words.

8 Is mutation always harmful? Explain your answer.

9 Are hydrogen bonds strong or weak? Explain your answer.

10 What type of compound is 'ribose'?

11 What is meant by 'deoxy'?



In this exercise you will study the inheritance of four characteristics, or traits, of some mice-like creatures. You will need to revise your knowledge of the following technical terms used in genetics: cross, pure-breeding, genotype, phenotype, offspring, F₁ generation, F₂ generation, homozygous, heterozygous, dominant, recessive, and Punnett squares.

The table below shows the selected traits.

Note: ♂ = male ♀ = female

Characteristic	Dominant			Recessive		
Tail shape	T		straight	t		curly
Ear shape	E		round	e		oval
Coat type	C		smooth	c		rough
Whisker length	W		long	w		short

Using Punnett squares, find the ratios of the genotypes and, hence, the phenotypes of the offspring of the crosses. Draw the appropriate traits (phenotypes) on the 'mouse' pictures.

Cross 1

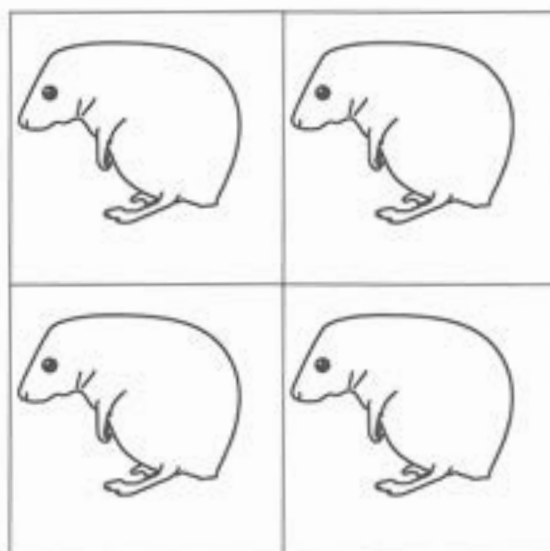
What are the genotypes and phenotypes of the F₁ generation resulting from a cross between a pure-breeding, straight tailed male and a pure-breeding, curly tailed female?

Genotypes (parents): ♂ _____ × ♀ _____



Genotypes of F₁: _____

Phenotypes of F₁: _____





Cross 2

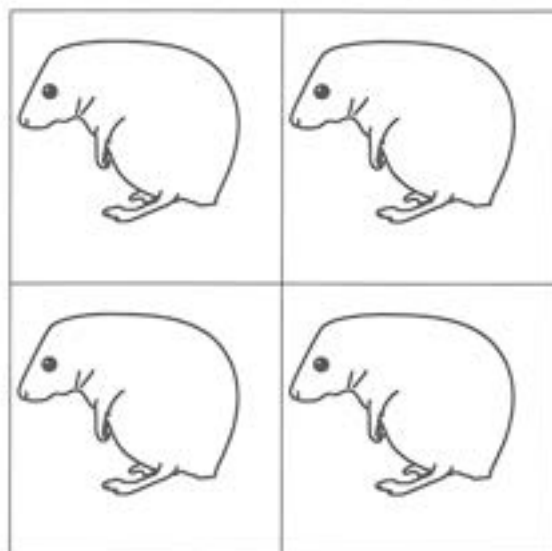
A male and a female from 'cross 1' were allowed to breed. What are the genotypes and phenotypes of the $F_1 \times F_1$ cross?

Genotypes (parents) F_1 : ♂ _____ $\times$ ♀ _____



Genotypes of F_2 : _____

Phenotypes of F_2 : _____



Cross 3

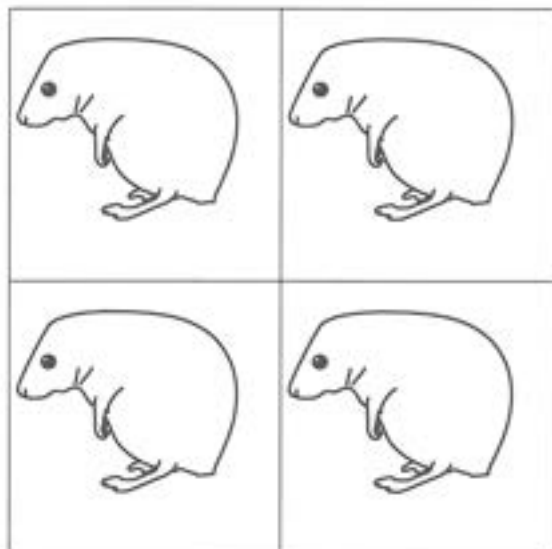
Consider a cross between a homozygous oval-eared male and a homozygous round-eared female. What are the genotypes and phenotypes of this cross?

Genotypes (parents): ♂ _____ $\times$ ♀ _____



Genotypes of F_1 : _____

Phenotypes of F_1 : _____



Cross 4

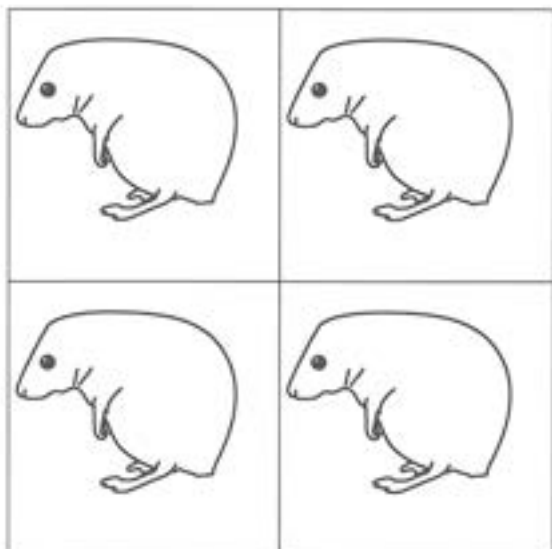
The F_1 offspring from 'cross 3' are allowed to breed. What are the possible genotypes and phenotypes of the F_2 generation?

Genotypes (parents) F_1 : ♂ _____ $\times$ ♀ _____



Genotypes of F_2 : _____

Phenotypes of F_2 : _____





Cross 5

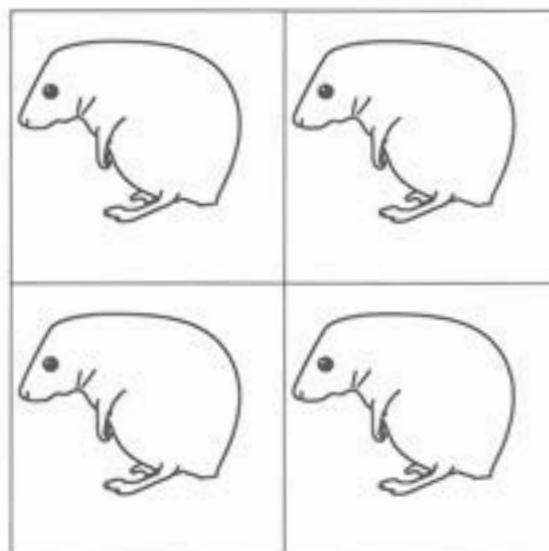
Consider a cross between a heterozygous long-whiskered male and a homozygous short-whiskered female. What are the genotypes and phenotypes of this cross?

Genotypes (parents): ♂ _____ × ♀ _____



Genotypes of F₁: _____

Phenotypes of F₁: _____



Cross 6

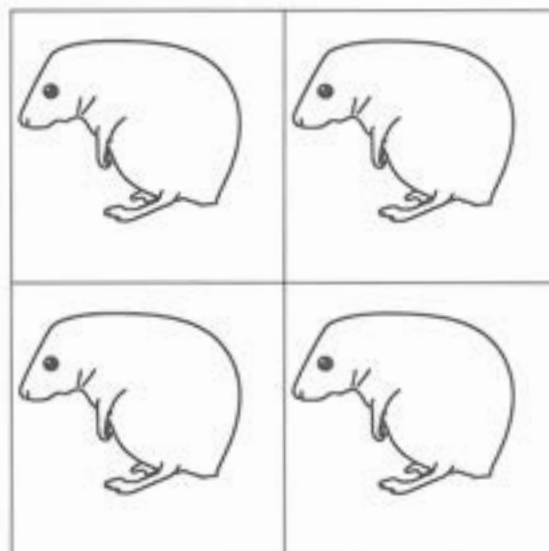
Consider a cross between a homozygous rough-coated male and a homozygous smooth-coated female. What are the genotypes and phenotypes of this cross?

Genotypes (parents): ♂ _____ × ♀ _____



Genotypes of F₁: _____

Phenotypes of F₁: _____



Cross 7

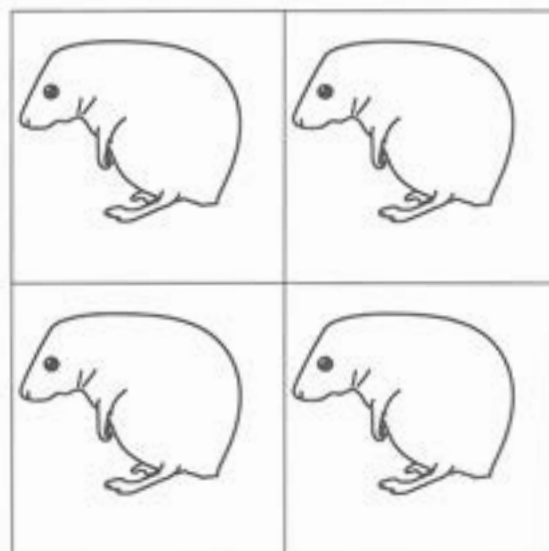
The offspring of 'cross 6' are allowed to interbreed. What are the genotypes and phenotypes of this cross?

Genotypes (parents) F₁: ♂ _____ × ♀ _____



Genotypes of F₂: _____

Phenotypes of F₂: _____



Dominant and recessive genes

In organisms which reproduce sexually there are two copies of every gene, one in each member of every chromosome pair (the only exception being the sex chromosomes of the male). Mendel first stated this in his 'Law of Segregation'. This says that there are two factors for each characteristic (trait) in an individual. When both copies of a particular gene have identical alleles, the individual is said to be *homozygous* for that gene. However, many paired genes will have different alleles and the individual is said to be *heterozygous* for such genes. One allele may be dominant in that the gene product for which it is the code is used by the cell in preference to that of the other allele, which is called *recessive*.

The outcome of such interactions between the alleles of all the different genes produces the observable characteristics of an individual, known collectively as the *phenotype*. The entire set of alleles of an individual is known as its *genotype*.

Monohybrid cross

This is the inheritance of one single characteristic. Mendel investigated the height of peas (among other traits) around 1865. Most people at that time thought that inheritance involved a type of 'blending'. This would mean that, if you crossed a tall organism with a short organism, for example, then the result would be offspring of a medium size. Mendel crossed tall plants with dwarf plants and found that all of the first generation of offspring (F_1) were tall. These offspring plants were then themselves crossed to give a second generation (F_2). These second generation plants were in the ratio of three tall to one dwarf plant.

Mendel repeated his experiments several times to make sure that there had been no errors. He came to the conclusion that tallness was a dominant characteristic, but the dwarf characteristic, which didn't disappear but came back in later generations, was recessive. If we call the gene for tall 'T' and the gene for dwarf 't', and we know that the very first plants were homozygous, then tall plants can be designated 'TT' and dwarf plants can be 'tt'.

If you cross homozygous tall (TT) plants with other homozygous tall (TT) plants then the offspring can only be homozygous tall (TT) plants. A similar principle applies if you cross homozygous dwarf (tt) plants with homozygous dwarf (tt) plants — you end up with homozygous dwarf (tt) plants.

But what happens if you cross tall plants with dwarf plants?

Phenotype: tall $\times$ dwarf

Genotype: TT $\times$ tt

Then, as shown in the 'Punnett square' on the right, the resulting (F_1) generation are all heterozygous tall (Tt) plants.

Note that, in Punnett squares, convention shows the male parent as the horizontal (top) component and the female parent as the vertical (side) component. We use a capital letter for the dominant allele.

	T	T
t	Tt	Tt
t	Tt	Tt

	T	t
T	TT	Tt
t	Tt	tt

What happens if we then cross the heterozygous (Tt) plants? The resulting (F_2) generation is half heterozygous tall (Tt) plants, one quarter homozygous tall (TT) plants and one quarter homozygous dwarf (tt) plants. In other word there will be three times as many tall plants as dwarf ones.

The same pattern is found in the inheritance of certain human characteristics, including recessive genetic diseases such as cystic fibrosis and sickle-cell anaemia. Women can carry a (recessive) 'haemophilia' gene without themselves being affected. Red-green colour blindness is inherited in a similar fashion. There are 4% of Australian males, but fewer than 0.5% of Australian females suffering from red-green colour blindness.

Huntington's chorea is a rare, inherited disease, but the gene responsible for this complaint is a dominant characteristic. This means only one allele 'dose' of the characteristic is needed for a person to be affected. Symptoms (clumsiness, twitching and nervous tics) usually occur when the affected persons are in their 40s and 50s, and are followed by death within 10 to 20 years. Genetic testing can identify adult carriers, or even embryos ('in utero') carrying the specific gene.

Research is being carried out into *gene-replacement therapy* (the replacement of defective genes with normal copies) to cure many inherited diseases. However success in this field is unlikely to be seen in the near future.

The environment can also influence growth and development of organisms, through the amount of food, heat, shelter and stimulation received.

- 1 What is the difference between 'dominant' alleles and 'recessive' alleles?

- 2 What do the genetic terms ' F_1 ' and ' F_2 ' represent?

- 3 Define 'genotype' and 'phenotype'. Expand your answers with reference to plants mentioned on Worksheet 40A.

- 4 One variety of mouse is called the 'Agouti' and has brown fur. Albino mice have white fur. Given that the genotypes for the two varieties are AA and aa respectively, show (using a Punnett square) what the results would be of crossing these two varieties, and then crossing the first generation of offspring. Write an explanation of the results.

- 5 Apart from heredity, what can affect the growth and development of an organism?

- 6 What is a 'monohybrid cross'?

- 7 Who was the first scientist to investigate a 'monohybrid cross'?

- 8 Name two inherited recessive genetic diseases. _____
- 9 Briefly state Mendel's 'Law of Segregation'. _____
- 10 Why might parents and children carrying the gene for Huntington's chorea not show symptoms for the disease?

- 11 If a parent is a carrier of the Huntington's chorea gene, does this automatically make all children carriers? Explain your answer.

Sexual dimorphism: occurs in a species when there are clear differences in appearance (in addition to the reproductive organs) between males and females.

Sexual selection: when 'survival of the fittest' is ignored

Sexual selection is often directed *not* by what will produce the fittest individual, but by female preference! The size of a male deer's antlers, a male lion's mane or the bright plumage on males of some bird species are examples of features affecting sexual selection among wildlife. Females often choose to mate with males displaying the most 'attractive' features or producing the most elaborate courtship/display rituals. Over time, because these features attract more mates, the feature becomes more apparent throughout populations.

Consider tail length in birds. Tail length in some species is seen as a desirable trait in males.

1 Suggest ways in which a longer tail could be harmful to individuals. _____

2 When features have evolved as a result of sexual selection, we might expect them to become even more elaborate. This does not happen. Give reasons for this.

3 Apart from physical attributes, suggest one way males can attract mates.

Sexual selection at work around us

Human males and females go to great lengths to attract mates. Many of these practices are of no benefit to the fitness of the species.

4 List examples of things done by human males and females to attract mates, that are of no benefit to their 'fitness to survive'. Discuss if any are harmful.

Darwin was aware of sexual selection, which he viewed as a separate selection process that caused increased differences in secondary sexual characteristics between males and females.

Sexual selection directs evolution by increasing a trait that usually reduces 'fitness to survive'.

5 Male deer develop antlers. In addition to attracting mates, how might males that have large antlers be advantaged?

Normal human body (somatic) cells contain 23 pairs of chromosomes. In 22 of the pairs, both chromosomes look the same as each other. In one pair, the chromosomes do not always resemble each other. In females, these chromosomes are the same, and are called X chromosomes. In males, one of the chromosomes looks the same as the X chromosome in the female cells. The other looks different and is called the Y chromosome. So, human females have 22 pairs of 'normal' chromosomes and an XX pair. Human males have 22 pairs of chromosomes and an XY pair. Extra sex chromosomes are found in about one in 1000 people. The three most common examples of this are XXY, XYY and XXX.



- Complete the Punnett square on the right to show what the ratios of sex (gender) genotypes in offspring will be between a human male and a human female.
- From your completed Punnett square, what is the ratio of males to females?

- How many pairs of chromosomes do humans have in somatic cells?

Genes are grouped together on chromosomes. Those found on X or Y chromosomes are said to be 'sex-linked'. The Y chromosome doesn't carry very many genes. It does carry the information to give male characteristics.

- Name two male secondary sexual characteristics. _____

In males, nearly all the genes on the X chromosome, whether or not they are dominant, will affect many functions, not necessarily related to sex. This means that there is only one form of most genes on the sex chromosomes of males.

Characteristics controlled by genes on the X chromosome are sex-linked because they occur more often in one sex than the other.

Another way of saying this is that the XX chromosomes in a female contain many paired genes. The Y chromosome in a male does not have partners for most of the genes on the X. So any recessive genes on the X chromosome will occur more often in males. These unpaired genes on the X chromosome, are sex-linked.

- What is a *recessive* gene? _____

If there is a defective gene on a part of the X chromosome of an XY pair, there sometimes isn't a corresponding gene on the Y portion. There is no 'backup' (such as a dominant allele) and symptoms will result. Over 50 inherited diseases are sex-linked. Examples of these are Duchenne Muscular Dystrophy (DMD), red-green colourblindness and haemophilia (a disease in which blood doesn't clot). The disease DMD is caused by a break in the X chromosome at the site of the DMD gene. Over 80% of haemophilia is family-linked, while other cases can be caused by spontaneous mutations.

A gene related to colour vision is found on the X chromosome, but not on the Y chromosome. This is a recessive gene.



Look at the possible offspring resulting from a mother who carries one gene for colour blindness (and will not be colour blind herself) and a 'normal' father who doesn't carry any genes for the disease.

If we represent the colour blindness gene as X^c and the normal gene as X , the mother is X^cX and the father is XY .

- 6 Complete the Punnett square on the right to show the genotypes of the offspring of the X^cX mother and the XY father.

- 7 From the Punnett square you completed, what are the proportions of the phenotypes of the offspring? State whether these are male or female.

- 8 What do we mean by a 'carrier' for a genetic disease?

Drosophila melanogaster (a type of fruit fly) are small two-winged flies also known as 'red-eyed pomace flies'. They are often used in genetics experiments because they breed quickly. Thomas Hunt Morgan (1866–1945) was an American geneticist, who carried out studies of heredity in *Drosophila* and was awarded a Nobel Prize in 1933 for discovering that genes for specific traits were located at specific sites on chromosomes. He experimented with eye colour.

He found that eye colour (red or white) is a sex-linked inheritance in *Drosophila*. Red eyes are dominant (R) over white eyes (r).

- 9 Using the Punnett square on the right, work out the possible genotypes of a cross between a white-eyed female (X^rX^r) and a red-eyed male (X^RY).

- 10 From the Punnett square, what are the proportions of the phenotypes in the offspring? State whether these are male or female.

- 11 Name an inherited disease, apart from colour blindness, that is sex linked. _____

- 12 Why do geneticists use fruit flies to carry out experiments?

- 13 What is the common name for *Drosophila melanogaster*? _____

- 14 Explain the difference between the way that inherited diseases that are 'sex-linked' are passed on and those that are due to recessive genes are passed on.

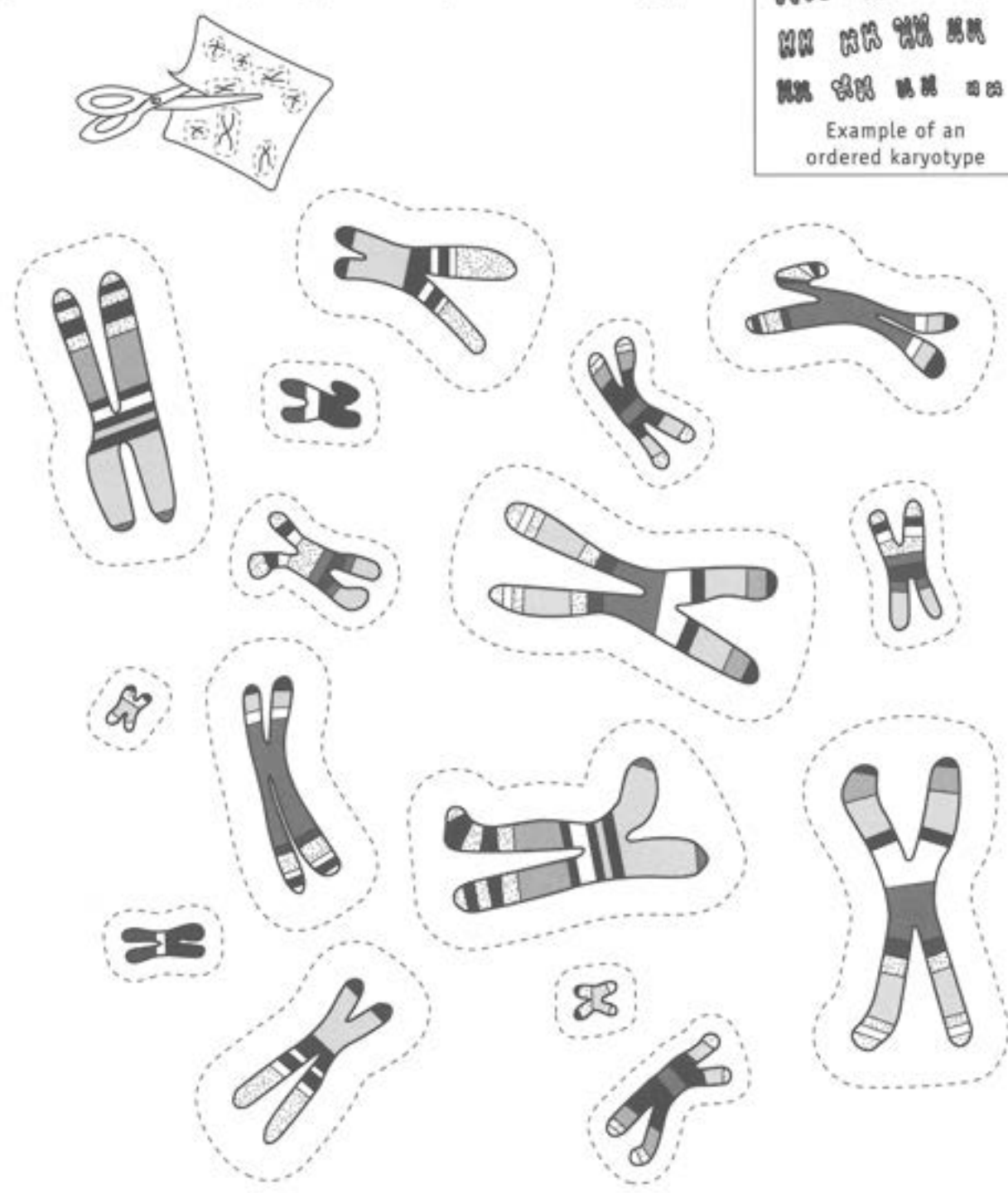
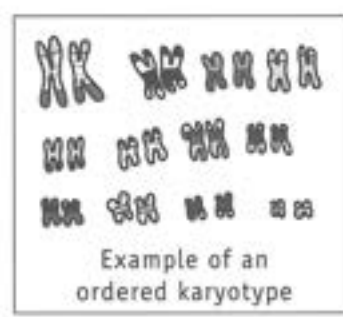
- 15 What are the symptoms of haemophilia? _____

- 16 'About one in a thousand people will have three sex chromosomes.' True or false? _____

Karyotype: a display of homologous chromosomes arranged in order of size and shape.

In order to study the chromosomes of an individual, geneticists often make karyotypes. To do this, chromosomes are extracted, stained and photographed. Each chromosome then shows a distinctive pattern of bands which are the key to pairing (shapes can vary slightly). The jumble of chromosomes must be sorted and paired — usually by entering the photograph into a computer and electronically rearranging it to illustrate the genetic make up of the individual being studied. The following exercise is a hypothetical 'cut and paste' version of this process.

- 1 Cut out the chromosomes below, using the dotted lines as a cutting guide.
- 2 Glue the chromosomes in pairs in your book to represent an ordered karyotype.





Drosophila melanogaster is popular (among geneticists) for several reasons, including the following:

- because it is so tiny, it is easy to house and feed — only minute amounts of cereal, honey or molasses, and agar are needed
- each female lays hundreds of eggs, most of which grow to maturity within 10–14 days
- mutants arise regularly, allowing geneticists to map the flies' genes thereby learning more about the field of genetics in general
- it only has four pairs of chromosomes, which are large and easily seen under a microscope.

The table on the right shows traits that have been found in fruit flies. The allele symbols are indicated in brackets.

Normal (wildtype ⁺): dominant	Mutant (sometimes shown as ⁻): recessive
five leg segments (L ⁺)	four leg segments (l ⁻)
grey body (B ⁺)	black body (b ⁻)
straight wings (W ⁺)	curly wings (W ^c)
	tiny, poorly functional wings (W ^h)
red eyes (E ⁺)	purple eyes (E ^p)
	white eyes (E ^w)
	blind/eyeless (E ^b)
long aristae* (A ⁺)	short or absent aristae (A ⁻)

* Aristae are bristly appendages resembling antennae on the front of the head.

Gene shuffling

Brother Mendel showed us how traits can be shuffled and reshuffled as they are passed from generation to generation. Study the cards on work sheet 44B. Cut them out. Shuffle them thoroughly and lay them face down. Turn over the cards one by one and take the first card for each trait as it appears. Continue this process until you have a card for each trait (leg segments, body, wings, eyes and arista) discarding any cards you turn over for a trait for which you have already selected.

- 1 What is the genotype of your fly?
- 2 What is the phenotype of your fly?
- 3 Cross your fly with one obtained by another student. Consider one trait at a time and draw Punnett squares to illustrate your cross.
- 4 Describe the genotype of the cross.
- 5 Describe the phenotype of the cross.
- 6 Back cross each parent with one of the offspring. Use Punnett squares to illustrate your crosses, one trait at a time. Then answer Questions 4 and 5 for your crosses.
- 7 How many generations of fruit flies could (theoretically) be studied in a year? Show your calculations.



Drosophila melanogaster: the geneticist's favourite pet

44B

Cards for 'gene shuffling'

Cut out these cards, and use them to obtain the traits of your individual fruit fly.

L ⁺	L ⁺	L ⁺	L ⁺	L ⁺	L ⁺
I ⁻	I ⁻	I ⁻	I ⁻	I ⁻	I ⁻
B ⁺	B ⁺	B ⁺	B ⁺	B ⁺	B ⁺
B ^b	B ^b	B ^b	B ^b	B ^b	B ^b
W ⁺	W ⁺	W ⁺	W ⁺	W ⁺	W ⁺
W ^c	W ^c	W ^c	W ^c	W ^c	W ^c
W ^s	W ^s	W ^s	W ^s	W ^s	W ^s
E ⁺	E ⁺	E ⁺	E ⁺	E ⁺	E ⁺
E ^p	E ^p	E ^p	E ^p	E ^p	E ^p
E ^w	E ^w	E ^w	E ^w	E ^w	E ^w
E ^b	E ^b	E ^b	E ^b	E ^b	E ^b
A ⁺	A ⁺	A ⁺	A ⁺	A ⁺	A ⁺
A ⁻	A ⁻	A ⁻	A ⁻	A ⁻	A ⁻



Genetically modified organisms (GMOs) are organisms that have had their genetic make-up altered by humans. It has been argued that, by selectively breeding plants and animals to suit our purposes, humans have been genetically modifying organisms for years. So why is there now so much concern?

Selective breeding involves choosing stock based on a characteristic we want to produce. Consider the field of beef production. Australian consumers like lean beef, whereas the Japanese market prefers 'marbled' beef (streaked with fat). Cattle breeders decide which type of beef they will produce and choose breeding stock based on animals possessing the characteristics (and genes) they require. This may involve cross-breeding species, which would not normally mate *but are still related*.

The GMOs, by contrast, are created by combining genetic materials from *completely unrelated species*. For example, when scientists announced efforts to produce grain crops able to thrive in drought-stricken areas, or resistant to pests, these were greeted with excitement. However, when it was explained that genes from the venomous Funnel Web spider had been inserted into one of the pest-resistant crops, the positive response turned to horror in some quarters. 'Frankenstein Foods' was one description of these crops. Public concern over the safety of 'playing with Mother Nature' in this way has led to severe restrictions on the production of genetically modified foods.

Selective breeding uses variations existing naturally among populations. However, GMOs often combine completely unrelated genetic materials that would never unite in nature.

For or against: what do you think?

The subject of GMOs will certainly continue to generate heated debate. List reasons for and against the production of GMOs. You may use examples other than those already mentioned.

- 1 The advantages of GMO production include:

- 2 The disadvantages of GMO production include:

- 3 Give two characteristics of sheep for which breeders may have selectively bred.

- 4 Compile a genetics scrap book by collecting newspaper and magazine articles. Comment on each article. Do you agree/disagree with the content? Why? Is the story linked to something else? Is it ethical?



Sex chromosomes determine the sex (gender) of an individual. A developing zygote receives one sex chromosome from each parent. An individual organism's sex is the result of a chance event. Males (♂) can give their offspring either the female X chromosome or the male Y chromosome. Females (♀) can only contribute an X chromosome. The X chromosome contains far more genetic instructions than the smaller Y chromosome, but both are important in an individual's development.

- 1 Name the two sex chromosomes a female carries.
- 2 Name the two sex chromosomes a male carries.
- 3 Which parent determines the sex of a child?
- 4 Explain your answer to Question 4.
- 5 What is a 'zygote'?



As a rule, one body plan is available to both sexes of most animals. How each animal develops is determined by the sex chromosomes it inherits. Each sex chromosome holds certain instructions determining which genes will be switched on or off in the developing baby. The presence of certain Y-associated genes acts as a trigger to *switch off* genes that would otherwise be activated in the presence of two X chromosomes.

- 6 The female body plan is more complex than the male body plan. Briefly explain this.

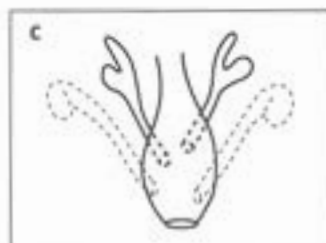
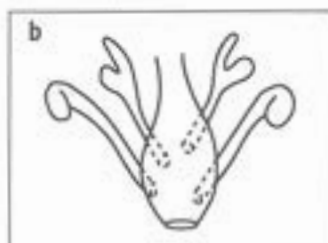
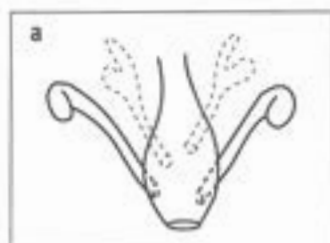
The mechanism using X chromosomes and Y chromosomes is not the only way that sex is determined. Many insects (such as ants and bees) don't have special sex chromosomes. Instead fertilised eggs will develop into (diploid) females and unfertilised eggs will develop into (haploid) males.

In the early stages of development, the material for both male and female reproductive systems are present. Rather than code every detail from scratch, it seems to be an easier process to have all basic parts available to be developed in whichever way is required, then to disintegrate the unnecessary organs. Many animals tend to have urogenital organs (that combine the functions of excretion and reproduction). There are two coexisting systems, the Wolffian duct and the Müllerian duct. If the individual is male, then the Wolffian duct eventually develops into the male system (penis, etc.) and the Müllerian duct becomes vestigial. If the individual is to become female then the Müllerian duct becomes predominant and the Wolffian duct degenerates. The external development of penis, vagina, etc. does not happen until internal development has determined the 'program' to be followed.

Copy and complete the following passage by adding the missing words.

Most higher mammals use X and Y (7) to decide the sex of their offspring. Each parent gives (8) sex chromosome to each offspring. In the presence of a (9) combination, genes are activated to begin developing the Wolffian ducts. At the same time the growing body is instructed to dissolve the unnecessary Müllerian (10). In the presence of a (11) combination, (12) are activated to begin developing the Müllerian ducts. In the presence of specific hormones and gene coding, the unnecessary (13) ducts aren't able to continue and so they wither and die.

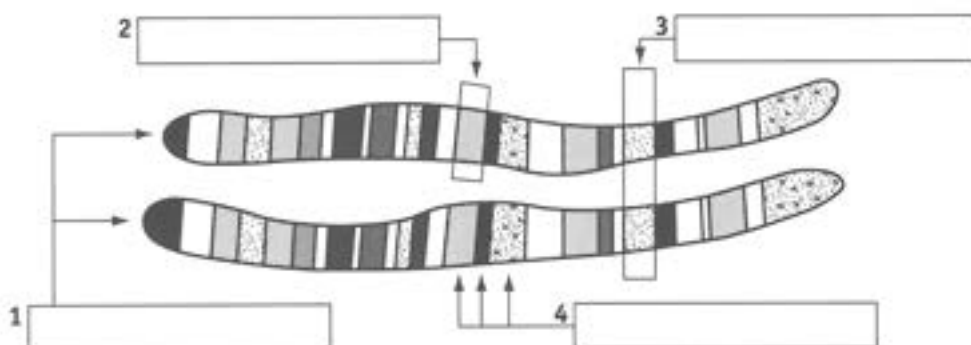
- 14 Label the following diagrams using 'Early stages of development', 'Developing male' or 'Developing female'. (The broken lines indicate disintegrating tissue.)



Label the parts (1 to 4) on this diagram using:

- allele
- allele pair
- gene locus
- homologous pair

Match the terms used in the chromosome diagram above with their correct definitions below:

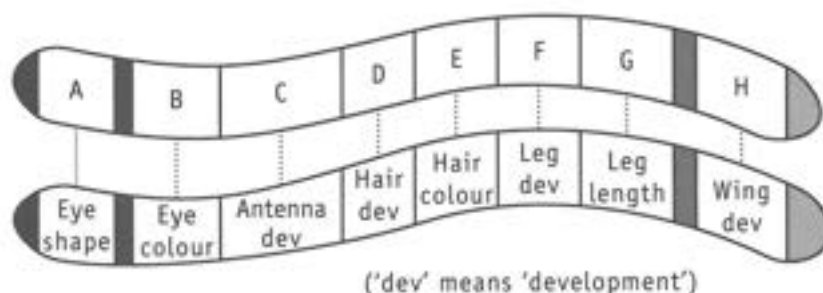


- 5 One half of a complete set is derived from each parent. _____
- 6 Pair of genes required to supply the code for a specific trait. _____
- 7 The exact location of a gene on a chromosome. _____
- 8 One of several forms of a gene for a characteristic. _____

Mapping

In the same way that cartographers identify and draw land masses when map-making, genetic maps can also be designed. As researchers discover which genes on which chromosomes are responsible for specific traits, the loci (plural of *locus*) are given numbers and a map is created. When homologous pairs align correctly, coding for a particular character occurs.

Study the hypothetical homologous pair below. The letters correspond to the traits described on the same locations on the other half of the pair.



- 9 Which locus on the diagram above is responsible for leg length? _____
- 10 Which locus is responsible for eye shape? _____
- 11 Which locus is responsible for antenna development? _____
- 12 Which locus is responsible for leg development? _____
- 13 Which trait would be affected if the alleles at locus B were not expressed? _____
- 14 Which trait would be affected if the alleles at locus E were not expressed? _____
- 15 Which trait would be affected if the alleles at locus D were not expressed? _____
- 16 Which trait would be affected if the alleles at locus H were not expressed? _____



Genethics: a term created by merging 'genetics' and 'ethics'. The work of scientists is important, but they must always consider the implications and ethics of their research.

1 What is meant by 'implications'? _____

2 What is meant by 'ethics'? _____

Not too many years ago, transplanting human organs was considered impossible. Now transplants are regularly performed in hospitals all over the world. Doctors have many (moral and practical) decisions to make, even if they have a donor organ that is a good match for the transplant recipient.

3 Suggest three things doctors might consider before carrying out the transplant of an organ.

The Human Genome Project

This project had its origins in the idea that scientists could identify and label each gene on each chromosome of the entire human body. By knowing the exact location of every gene, scientists argue that we can eventually improve the quality of our lives.

4 Give three ways that the Human Genome Project might be used to help mankind.

It has been argued that knowledge gained from this project could be used in improper or unethical ways.

5 Give three ways that the Human Genome Project could be used, in your opinion, improperly.

Gene therapy

This involves the modification of existing genes, usually by replacing defective genes with functional ones.

6 Give two advantages of gene therapy.

7 Give two disadvantages of gene therapy.

8 Name two diseases caused by defective genes. Include the symptoms of each disease.



'Designer babies'

Scientists can now manipulate certain aspects of the development of a human baby. This could eventually mean that parents may choose the appearance and other features of their future child.

- 9 Suggest two possible advantages of this technology.

- 10 Suggest two possible disadvantages of this technology.

'Test tube babies'— In-vitro fertilisation

Despite the name, 'test tube babies' are not really created in test tubes.

- 11 Find out how 'test tube' embryos come about and how the mother takes over.

- 12 Why are 'test tube babies' brought into being?

'Test tube babies' are no longer a new issue but there are still people who oppose this technique.

- 13 Give two possible concerns that people might express about 'test tube babies'.

Cloning

This process involves the repeated division of a single parent cell to produce a duplicate organism. It has been suggested that human body parts and animal species (extinct and existing) should be cloned.

- 14 Suggest two possible advantages of this technology.

- 15 Suggest two possible disadvantages of this technology.

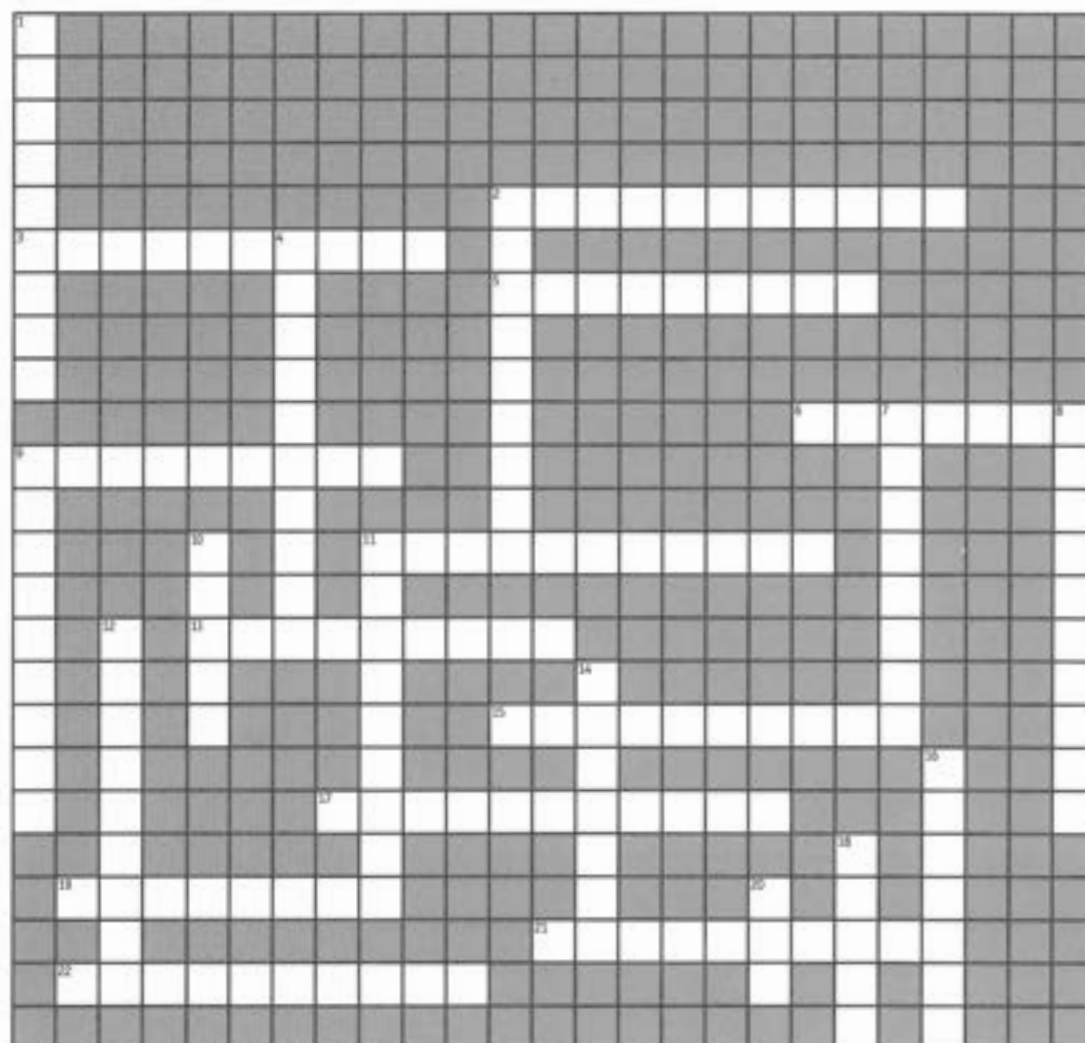
Stem cell research

This research shows great promise, but it is a very vigorously debated topic. There are concerns as to whether it is a 'good' thing to do.

- 16 Describe what is meant by a 'stem cell'?

- 17 Suggest two possible benefits that might result from stem cell research.

- 18 Why is this area of research so hotly debated among scientists and society in general?



Clues Across:

- 2 The process of making copies (11)
- 3 Chromosomes that look like each other, for example (10)
- 5 Joined, parallel strands of duplicated chromosomes (9)
- 6 Not sex cells, but body cells (7)
- 9 Where essential chemical life processes occur in the cell (9)
- 11 Also known as the 'centromere', where spindle fibres affect the movement of the chromosome (11)
- 13 Two minute, rod-shaped bodies concerned with forming the spindle (9)
- 15 A number of individuals of the same species in a location (10)
- 17 Division of cytoplasm (11)
- 19 Phenotype or genotype most common in a natural population (8)
- 21 Dependent on genes that follow specifically the X or Y chromosome (two words) (10)
- 22 Male parent first cross and female parent second cross, etc. (10)

Clues Down:

- 1 Final stage in mitosis (seen twice in meiosis) (9)
- 2 The alternative to dominant (9)
- 4 Can be F_1 , F_2 , etc. (10)
- 7 A sudden change causing genetic variation (8)
- 8 This carries coded hereditary information (10)
- 9 Thread-like mass of nucleic and basic proteins (9)
- 10 Where a gene or allele is located on a chromosome (5)
- 11 Ordered display of an individual's chromosomes (9)
- 12 Not random, but chosen (9)
- 14 Long protein fibres forming during the metaphase (7)
- 16 Structure essential for coordination of all cellular activity (7)
- 18 A characteristic that is inherited (5)
- 20 Nucleic acid that is an essential component of chromosomes (3)



The Hardy-Weinberg Law

50A

It should come as no surprise that the two men responsible for this law are named Hardy and Weinberg. In 1908, Godfrey Hardy (an English mathematician) and Wilhelm Weinberg (a German Doctor) independently described how, under certain conditions, predictions can be made about genotypes that will arise in a population.

The Hardy-Weinberg Law assumes that:

- the breeding population is sufficiently large
- other populations of the same species don't interbreed with the population studied
- mating is random
- all individuals are equally fertile and physically able to breed
- no mutation is occurring.

The ratios of alleles in the gene pool of a population will not be changed by sexual recombination, but only by outside influences.

- 1 Comment on one of the above assumptions of the Hardy-Weinberg Law.

A Punnett square can be used to calculate the inheritance of a single trait. It shows us how heterozygous crosses produce 75% of offspring carrying the dominant allele and 75% carrying the recessive allele. These figures might seem impossible, they add up to 150%, but are correct.

- 2 In the space provided on the right, draw a Punnett square showing the cross $Aa \times Aa$.

- 3 Use your results to explain why the statement '75% of offspring carry the dominant allele and 75% carry the recessive allele' is true. _____

Another tool, which can be used for more than one trait, is the equation:

In this equation p represents the dominant allele, q represents the recessive allele and 1 stands for 100% of the population.

$$p + q = 1$$

Drosophila melanogaster is a species of fruit fly. The species is infamous in Australia for causing damage to fruit crops but, for many years it has been an invaluable tool for scientists studying genetics. The main reason is that a *Drosophila melanogaster* individual can be born, live, breed and die within a period as short as two weeks. These fruit flies also have a considerable number of genes in common with humans.

- 4 Explain why the short life cycle of *Drosophila melanogaster* makes it a favourite with geneticists.

Example

Imagine we are working with a population of 100 *Drosophila melanogaster* in a jar. This particular population contains the following genotypes: 30 AA, 60 Aa and 10 aa.

The proportion of dominant alleles (A) is calculated by substituting into the equation, $p + q = 1$, $p = 30 \text{ AA} + (0.5 \times 60) \text{ Aa} = 60$. Note: we use 0.5 Aa because only one allele in the pair is the dominant allele.

The proportion of recessive alleles (a) is calculated as follows: $q = 10 \text{ aa} + (0.5 \times 60) \text{ Aa} = 40$.

Now check. Does $p + q = 1$?

Substitute values: $60 + 40 = 100$ (or 100% of the population studied)

We have accounted for the complete population, so $p + q = 1$.



The Hardy-Weinberg Law

50B

Time to do some sums!

Suppose that jar 'A' contains fruit flies with these genotypes: 29 AA, 44 Aa and 16 aa.

- 5 What is the total number of fruit flies in jar 'A'? _____
- 6 Using the formula given on work sheet 50A, calculate p . Show your working.

Often, before embarking on a calculation, we make an estimate to gain a rough idea of the answer.

- 7 Do some quick mathematics to estimate what you expect q to be.

- 8 Now calculate q , showing your working. _____

Jar 'B' contains fruit flies with the following genotypes: 20 AA, 18 Aa and 7 aa.

- 9 What is the total number of fruit flies in jar 'B'? _____
- 10 Calculate p , showing your working. _____

- 11 Calculate q , showing your working. _____

Minding your ps and qs !

If you use a Punnett square to cross pq and pq , you will find that:

- $\frac{1}{4}$ of offspring are homozygous dominant (pp)
- $\frac{1}{2}$ of offspring are heterozygous (pq)
- $\frac{1}{4}$ of offspring are heterozygous recessive (qq).

Another way to express this mathematically is shown on the right.

This is the same as saying $(AA)^2 + 2 Aa + (aa)^2 = 1$

$$p^2 + 2pq + q^2 = 1$$

where 1 = 100%

- 12 Find the number that represents pp (or p^2) for jar 'A' above. _____
- 13 Find the number that represents $2pq$ for jar 'A' above. _____
- 14 Find the number that represents qq (or q^2) for jar 'A' above. _____
- 15 Find the number that represents pp (or p^2) for jar 'B' above. _____
- 16 Find the number that represents $2pq$ for jar 'B' above. _____
- 17 Find the number that represents qq (or q^2) for jar 'B' above. _____
- 18 What is meant by 'homozygous dominant'? _____

- 19 What is meant by 'heterozygous'? _____

- 20 What is meant by 'homozygous recessive'? _____

- 21 'Fruit flies (*Drosophila melanogaster*) have approximately 14 000 genes.' True or false? _____

Genetics and environment

The environment in which an organism lives can affect how its genetic information is expressed. The Siamese cat has a pale coat arising from a mutation in the pigmentation gene. Pigmentation produces an organism's colour. The Siamese mutation (selected for by breeders because of the distinct appearance it produces) results in the chocolate colour only being expressed at the coolest parts of the body. On cats, these areas are the face, ears, paws and tail.



- 1 Give another example of how an organism's environment can affect its phenotype.

- 2 Some animals (including humans) have defective pigmentation genes, producing white hair, pale skin and red eyes with a blue or pink iris. Name this condition. _____

Random X-inactivation

Another easily recognisable cat is the Calico cat, which has a 'patchwork' coat of yellow, black and white. With the exception of identical twins, no two Calicos will be alike because their colouration is randomly decided during early development. Fur colour on cats is determined by genes on the X chromosome and the patchwork results from a process called *random X-inactivation*. As the name suggests, certain instructions on the X chromosome are inactivated at random.

Female cats receive two X chromosomes (males receive one). In Calico cats, one chromosome will contain the black gene (X^b) and the other contains the (alternative) yellow gene (X^y). As cells divide during development, one X chromosome in each cell will be inactivated at random. Daughter cells arising from the original parent cell will possess instructions to grow fur of the remaining colour. Sometimes the genes are instructed not to produce any colour and this is where the white fur appears. (Note: linkages to other genes are involved in this.)

- 3 Is it possible to produce a breed of Calico cat in which individuals are almost identical? Explain your answer.

- 4 Yellow 'Marmalade' cats are more commonly male than female. Why?

- 5 'Mr Big' was an enormous black and yellow cat. Was this cat male or female? Explain.

- 6 Is it possible to have male cats that are black and white, or yellow and white? Explain.

- 7 Why might cat breeders try to change the appearance of cats?



Coming to a Conclusion

52

Each statement given below is correct, however not all of the given conclusions drawn from them are correct. Decide whether each *conclusion* is true or false and explain the reason for your answer.

Statement: Nucleotides code for amino acids.

Amino acids code for proteins.

Conclusion: Nucleotides are proteins.

1 True or false? _____ Reason: _____

Statement: Recessive genes usually need to be present in double doses to be expressed.

When a lethal gene is present, an organism will die.

Conclusion: A single dose of a recessive lethal gene is enough to kill an organism.

2 True or false? _____ Reason: _____

Statement: Selective breeding can lead to negative side-effects.

Certain favourable traits are genetically linked to negative traits.

Conclusion: Selective breeding always produces an organism that is less fit for survival.

3 True or false? _____ Reason: _____

Statement: Rare forms of alleles tend to be found in single doses.

Selective breeding for specific traits in dogs often involves breeding animals with rare alleles.

Conclusion: Selective breeding can lead to an unnaturally high increase in the frequency of recessive alleles within a species' gene pool.

4 True or false? _____ Reason: _____

Statement: In the DNA double helix, adenine always bonds with thymine.

In the DNA double helix, cytosine always bonds with guanine.

Conclusion: In DNA molecules there will always be equal amounts of adenine and guanine.

5 True or false? _____ Reason: _____

Statement: The four bases forming DNA are made of ring structures called purines and pyrimidines

Purines are larger molecules than pyrimidines.

Conclusion: To create the ladder shape of DNA, a purine must bond with a pyrimidine.

6 True or false? _____ Reason: _____



Find the answers to the clues below. The first letters of your answers to the clues (1 to 10) will reveal the answer to the mystery question (11).

- 1 One of two replicated chromosomes *while they remain attached by a centromere*.
- 2 Passing genetic information from a parent to offspring is known as this.
- 3 A strand of nucleotides containing adenine, cytosine, guanine and uracil.
- 4 Another word for 'progeny'.
- 5 Method of cell division in which identical daughter cells are produced.
- 6 Common blood types are A, B, AB and what?
- 7 Method of reproduction in which a male and a female each contribute half of the offspring's genetic makeup.
- 8 Period at which the female germ cell is released from the ovary.
- 9 Method of cell division in which genetically different daughter cells are produced.
- 10 Gradual changes in a population's gene pool take place by a process called what?
- 11 I hold a wealth of information. What am I? _____

Pairs

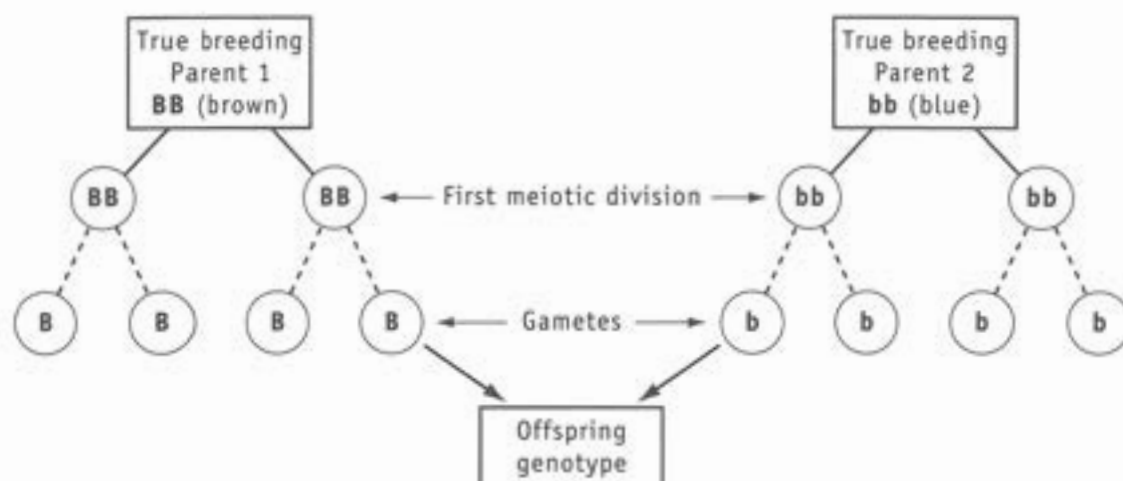
Select the correct term from the pair provided each time.

- 12 (*Somatic, Germ*) cells develop into body parts.
- 13 To be (*hybrid, pure-bred*) an offspring's parents must be homozygous for certain traits.
- 14 Sperm is an example of a (*somatic, germ*) cell.
- 15 The (*locus, locum*) refers to a position on a chromosome.
- 16 (*Hybrids, Pure breeds*) are the progeny of parents which are heterozygous for a certain trait.
- 17 (*Genotype, Phenotype*) refers to an individual's complete genetic make-up, whether or not it is expressed.
- 18 If two parents that are true-breeding for different traits BB and bb mate the resulting (F_1 , F_2) generation will always be (*homozygous, heterozygous*). Such a cross is referred to as a (*monohybrid, dihybrid*) cross.

Correct the spelling errors

Rewrite each of these genetics terms correctly.

- | | |
|--------------|-------------|
| 19 Fillial | 24 Nuclular |
| 20 Heridity | 25 Hibrid |
| 21 Myosis | 26 Someatic |
| 22 Feenotype | 27 Mytoses |
| 23 Cromasome | 28 Aleele |



The diagram above shows a cross between two 'true breeding' parents.

- 1 From the diagram, which parent is homozygous dominant? _____
- 2 Which parent is homozygous recessive? _____
- 3 The offspring's genotype would be referred to as what? _____
- 4 Is there any other outcome possible for the offspring? Explain your answer. _____

- 5 If the offspring is back-crossed with parent 1, work out (using a Punnett square) the possible outcomes. Include descriptions of the resulting phenotypes.

- 6 If the offspring is back-crossed with parent 2, work out (using a Punnett square) the possible outcomes. Include descriptions of the resulting phenotypes.

- 7 Explain why the two back crosses in Questions 6 and 7 produced different outcomes.



Blood is a vital body fluid containing plasma, platelets, red blood cells and white blood cells. In 1901, Karl Landsteiner formulated a blood classification system and also discovered that blood groups were inherited in a way that agreed with Mendel's laws of inheritance. In the ABO system, blood is classified depending upon whether the antigens A or B are present in red blood cells. Antigens are complex chemical substances.



Antigens: substances that cause the production of antibodies (defence proteins) to combat infection.

Group A blood has the A antigen, group B has the B antigen, group AB has both and group O has neither. This is because the gene for blood type has three alleles: I^A , I^B and I^O .

1 What is an 'allele'? _____

The genes I^A and I^B are dominant over I^O . However gene I^A is only incompletely dominant over I^B .

Because there are only two chromosomes of each type per cell, there can be only two alleles present at any one time.

The diagram on the right shows the blood groups that can be formed.

Genes	Blood type
$I^A + I^A$	A
$I^A + I^O$	A
$I^O + I^O$	O
$I^B + I^B$	B
$I^A + I^B$	AB
$I^B + I^O$	B

2 How many different blood groups are there according to this system?

3 Use the Punnett square on the right to show the possible blood groups of offspring, inherited from an $I^A I^B$ mother and an $I^B I^O$ father.

4 Use the Punnett square on the right to show the possible blood groups of offspring, inherited from an $I^A I^A$ mother and an $I^A I^B$ father.

5 Who was the person who first classified blood types as A, B, AB and O?

6 Suggest two examples of when blood types might be used in legal cases.



Chromosomes are clearly seen under a standard light microscope (that is one that uses glass lenses) when a cell is undergoing division. (Note: the cell must be stained first). Chromosomes get their names from two Greek words, *chroma* (meaning colour) and *somes* (meaning body), because they absorb certain dyes.

Cytology: a branch of biology that deals with the structure and function of cells.

Cytologists had a good idea of the sequences involved in mitosis and meiosis over a hundred years ago.

In the early 1900s biologists began to see that there was a link between the way chromosomes behaved and (the newly rediscovered) Mendel's 'factors'.

Walter Stanborough Sutton was an American physician and geneticist. He (and others) pointed out, in 1902, that Mendel's hereditary factors (genes) occurred in pairs in somatic cells and separated when gametes were formed (gametogenesis). This tied in with the fact that somatic cells contain two of each kind of chromosome and that these chromosomes separate during meiosis. After further study of meiosis (in grasshoppers) he observed that:

- in a cell undergoing meiosis, the chromosomes line up in pairs
- the chromosomes in each pair resemble each other strongly in physical appearance
- at one stage the chromosome pairs separate, giving each resultant gamete (sex cell) only one chromosome from the pair
- when a male sex cell and a female sex cell unite, the resulting zygote has a full complement of chromosomes.

Around the same time, Theodor Heinrich Boveri, a German zoologist, independently demonstrated the separate and continuous existence of chromosomes. Boveri is also noted for his discovery of the centrosome. Between the work of Sutton and Boveri (and other researchers) a chromosome theory of inheritance started to take shape. The points below summarise Sutton and Boveri's views at the time.

- Chromosomes carry the units of inheritance.
- Chromosomes occur in pairs.
- Organisms have many more units of inheritance than there are chromosomes.
- Each chromosome carries hundreds of inheritance factors (genes).

Centrosome: the region of clear cytoplasm, near the cell nucleus, that contains centrioles.

Wilhelm Ludvig Johannsen was a Danish botanist and geneticist who came up with the term 'gene' (around 1910) for a unit of heredity. He also was responsible for the distinction of genotype and phenotype.

Further work by Thomas Hunt Morgan (an American embryologist) with *Drosophila melanogaster* (fruit flies) confirmed Mendelism and the chromosome theory, and also led to the conception of the gene as the smallest unit of recombination.

- 1 What does a *cytologist* study?
- 2 What is 'gametogenesis'?
- 3 What is the origin of the name 'chromosome'?
- 4 Why is the chromosomal theory of inheritance referred to as the 'Sutton-Boveri' theory?
- 5 What do we now call Mendel's inheritance 'factors'?
- 6 Outline the views of Sutton and Boveri on the behaviour of chromosomes in meiosis.
- 7 Which organism was used by T. H. Morgan in his studies?



Identify the person (Questions 1 to 10) or the term (Questions 11 to 20) from the clues provided. Choose from the answers provided in this box:

pea	gene	C. Darwin	T. H. Boveri	chromosome	K. Landsteiner	J. Watson
fruit fly	sperm	W. S. Sutton	Gregor Mendel	nucleotide	A. R. Wallace	nucleus
RNA	allele	W. Johannsen	R. C. Punnett	W. Bateson	DNA	

- 1 I am a monk and my research in peas led me to theories of inheritance. _____
- 2 I am an American. Working with grasshoppers I provided the first conclusive evidence that chromosomes carry units of inheritance and occur in distinct pairs. _____
- 3 I made similar discoveries to W. S. Sutton and at the same time. I also discovered the centrosome. _____
- 4 I came up with a blood classification system at the beginning of the 20th century. _____
- 5 I was the first to distinguish between genotype and phenotype and coined the term 'gene'. _____
- 6 My squares help to decide genotypes and phenotypes of offspring. _____
- 7 With Francis Crick and Maurice Wilkins I received a Nobel prize for work on DNA. _____
- 8 I came to similar conclusions as Charles Darwin and had an imaginary geographical line named after me. _____
- 9 My first names are Charles Robert and I was a naturalist on the 'Beagle'. _____
- 10 I am a British biologist and I came up with the term 'genetics' in 1906. _____
- 11 You can only see me when I'm dividing and if you dye me. _____
- 12 Farmers consider me a nuisance but, in my small way, I have helped geneticists. _____
- 13 I am a common or garden legume but a lot of interest was shown in my inheritance. _____
- 14 I am a male gamete. _____
- 15 I am a large, double-stranded helical molecule. _____
- 16 I am only a small part of a chromosome, but I give the instructions. _____
- 17 I am one half of a pair of non-identical genes for a particular trait. _____
- 18 I am the centre of things in almost every cell. _____
- 19 I contain a sugar, a phosphate and a base group containing nitrogen. _____
- 20 I am similar to DNA, and I am also found in the cytoplasm. _____



More Genetic Terms

58

Match the term with the correct definition/description by putting the letter of the individual description in the box next to the word it describes.

	Term		Description
1	centriole	a	Ordered displays of an individual's chromosomes.
2	chromosome	b	Regions of centromeres where microtubules attach.
3	chromatids	c	Two bodies just outside the nucleus, vital to cell division.
4	chromatin	d	The location on a chromosome of a gene or allele.
5	cytokinesis	e	A change in hereditary material.
6	cytoplasm	f	Body which carries the coded, hereditary information.
7	DNA	g	The cell's control centre.
8	F_1/F_2	h	Total number of individuals of the same species in a location.
9	Hardy	i	Produces no effect when present with a contrasting allele.
10	homologous	j	Connected with Weinberg in a law concerning heritance.
11	karyotype	k	For example male parent first cross and female parent second cross, etc.
12	kinetochore	l	Making copies.
13	locus	m	Connected specifically to either the X or the Y chromosome.
14	mutation	n	Fibre network along which chromosomes arrange in division.
15	nucleus	o	The cytoplasm cleaving into two daughter cells.
16	population	p	Final stage of mitosis, the spindle disappears.
17	recessive	q	An inherited characteristic.
18	reciprocal	r	To do with the body.
19	replication	s	Phenotype most common in natural populations.
20	selective	t	Cell's material where chemical reactions vital to life occur.
21	sex linkage	u	Choosing or being specific.
22	somatic	v	Having the same alleles for a characteristic.
23	spindle	w	Nucleic acid that is the basis of hereditary.
24	telophase	x	Thread-like mass of nucleic acids and basic proteins.
25	trait	y	First and second generations.
26	wild type	z	Joined, parallel strands of duplicated chromosome.



Famous double acts

The following scientists are often named together when we talk of specific discoveries in genetics. For each pair of scientists:

- state the area of research/discovery with which they are associated
- state the time and the place in which they carried out their work
- state one other fact about each scientist.

1 Crick and Watson: _____

2 Sutton and Boveri: _____

3 Darwin and Wallace: _____

4 Hardy and Weinberg: _____

5 Beadle and Tatum: _____

Definitions

Give a brief definition/description of each of the following terms:

6 autosomal _____

7 polypeptide _____

8 evolutionary divergence _____

9 palaeontology _____

10 selective breeding _____



Fertility: seedless watermelons and the mule

60

The demands of the marketplace have led plant producers to modify existing plants (using selective breeding) to come up with something their customers will like. Many people like **fruit** without seeds. Seedless watermelons are produced by crossing a tetraploid ($4n$) melon with a **diploid** ($2n$). The resultant fruit (a triploid, $3n$) is mature but has low, or non-existent, **fertility**. As a result of this, to get more seedless melons, the crosses must continue because it is impossible to produce a seedless **species** with which to grow new plants. Many **hybrids** are used in food production, for example tomatoes, corn, rice, wheat, etc. Reproduction is often by grafting or cuttings. Sometimes fertility can be restored by treating with radiation, heat or chemicals.

Another hybrid, this time an animal, is the mule. It is a **domestic** working animal produced by crossing two closely related species, horses (*Equus caballus*) with donkeys (*Equus asinus*). Strictly speaking the mule is the offspring of a **mare** and a male donkey. The result of a cross between a female donkey and a **stallion** is called a hinny. Male mules are usually **sterile** but females have been crossed with both horses and donkeys to produce **foals**.

Sometimes selective breeding can produce a species that has poor reproductive potential and will fail to survive, without **intervention**.

Reread the passage above and explain, in the spaces provided below, what is meant by the words in **bold** type.

- 1 fruit _____
- 2 diploid _____
- 3 fertility _____
- 4 species _____
- 5 hybrids _____
- 6 domestic _____
- 7 mare _____
- 8 stallion _____
- 9 sterile _____
- 10 foals _____
- 11 intervention _____
- 12 Sometimes people refer to 'hybrid vigour' or *heterosis* when referring to traits exhibited by crosses of animals or plants. What is meant by 'hybrid vigour'? _____

- 13 Why are mules bred? _____

- 14 What is another name for an 'ass'? _____
- 15 Name another fruit (apart from the watermelon) that is grown without seeds. _____
- 16 What is the cause of sterility in hybrids such as the mule? _____



Questions 1 to 16 are multiple choice. Choose the response which best answers the question each time.

- 1** The abbreviation DNA stands for:
- A** deribose nucleic acid **B** deoxyribonucleic acid
C deolytic nucleic acid **D** denatured nucleic acid
- 2** An agent that causes a gene to change suddenly is called a:
- A** mutation **B** mutagen **C** mute **D** mutant
- 3** Alternative forms of a gene are called:
- A** algae **B** allegros **C** allergens **D** alleles
- 4** Organisms that undergo sexual reproduction normally have how many copies of every gene?
- A** 1 **B** 2 **C** 3 **D** 4
- 5** The term for an organism's *complete genetic makeup*, whether or not certain genes are expressed, is called its:
- A** phenotype **B** phenomenon **C** genotype **D** geniture
- 6** Which of the following statements is *false*?
- A** Chromosomes are not always clearly visible in cells.
B Genes are made of DNA.
C Most mutations are harmful to an organism.
D Chromosomes are found on genes.
- 7** Organisms produced by repeatedly dividing the cells of a single parent are called:
- A** clones **B** recombinants **C** codominants **D** codons
- 8** The resting phase of nuclear division and cell duplication is called:
- A** prophase **B** telophase **C** interphase **D** metaphase
- 9** How many pairs of chromosomes are found in human body cells?
- A** 12 **B** 23 **C** 32 **D** 46
- 10** Which of the following *does not* occur during mitosis?
- A** chromosome duplication
B cytoplasmic division
C chromosomes lining up on the cell's 'equator'
D crossing over



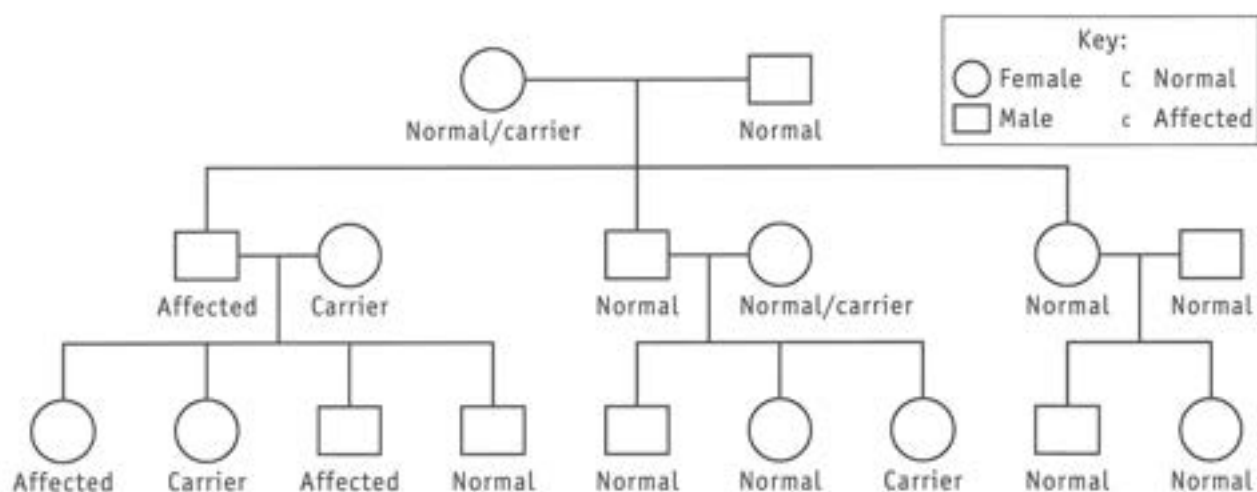
- 11** Which of the following statements is *true*?
- A** Meiosis produces daughter cells that are genetically identical to the parent cell.
 - B** Meiosis occurs in asexually reproducing organisms.
 - C** Mitosis produces daughter cells that are genetically identical to the parent cell.
 - D** Mitosis produces daughter cells that are genetically different to the parent cell.
- 12** Which of the following correctly summarises the content of DNA?
- A** sugar, phosphate and nitrogen-based nucleotides
 - B** sugar, carbonate and nitrogen-based nucleotides
 - C** sugar, nitrate and phosphorus-based nucleotides
 - D** sugar, nitrate and oxygen-based nucleotides
- 13** Which is the correct sequence of events in mitosis, from first to last?
- A** telophase, anaphase, metaphase and prophase
 - B** anaphase, metaphase, prophase and telophase
 - C** metaphase, prophase, anaphase and telophase
 - D** prophase, metaphase, anaphase and telophase
- 14** The four nitrogen bases in DNA are:
- A** alanine, cytosine, guanine and thymine
 - B** adenine, cytosine, glycerine and thymine
 - C** adenine, cytosine, thymine and guanine
 - D** adenine, cytosol, thiamine and guanine
- 15** An example of 'sexual selection' is when:
- A** parents choose the sex of their baby
 - B** males have elaborate features to attract females
 - C** more females than males survive in a particular environment
 - D** more males than females survive in a particular environment.
- 16** Who was the first researcher credited with using the fruit fly for genetic research?
- A** Thomas Hunt Morgan **B** Gregor Mendel **C** Walter S. Sutton **D** Theodore H. Boveri

Answer true or false for Questions 17 to 22.

- 17** 'Karyotypes are arrangements of homologous chromosomes of an organism.' True or false? _____
- 18** 'Wild-type alleles are the *normal* allele of an organism.' True or false? _____
- 19** 'The ring structures forming DNA are called purines and pyrites.' True or false? _____
- 20** 'In the DNA double helix, cytosine always forms bonds with adenine.' True or false? _____
- 21** 'Filial refers to sons and daughters.' True or false? _____
- 22** 'Humans have more chromosomes than any other organism.' True or false? _____



Questions 23 to 25 refer to the following information and diagram relating to a sex-linked condition.



- 23 In the spaces provided on the diagram, write the genotype of each individual shown.
- 24 What do you notice about the normal carriers? _____
- 25 Why are more males affected? _____
- 26 Briefly explain the difference between *contracting* a disease and *inheriting* one. Give two examples of each type of disease.
- _____
- _____
- _____
- _____
- _____
- 27 Explain why genetic variation is important among breeding populations.
- _____
- _____
- _____
- _____
- _____
- 28 Cloning human cells is a controversial issue. Give two arguments for and two arguments against cloning humans.
- _____
- _____
- _____
- _____



Answers

32 More About Mutations: changing DNA

- 1 Teacher to check answers. Answers should be logical.
- 2 Inversion (Region A).
- 3 Deletion (Region B).
- 4 Addition (Region B).
- 5 Translocation (Region C).

33 Recognising Mutations

Teacher to check that students have altered the original chromosomes in the manner described each time.

34 Genetics Review:

- 1 Mendel.
- 2 peas.
- 3 frequencies.
- 4 naturalist.
- 5 heredity.
- 6 Selection.
- 7 half.
- 8 nucleus.
- 9 complex.
- 10 dominant.
- 11 visible.
- 12 recessive.
- 13 absence.
- 14 Phenotype.
- 15 genotype.
- 16 molecular.
- 17 Population.
- 18 Mendelian.

35 Genetics Review: terms to know

- 1 Male yellow peas $\times$ female green peas
- 2 It is simpler and shorter.
- 3 Mitosis.
- 4 Meiosis.
- 5 Gamete.
- 6 Somatic.
- 7 Zygote.
- 8 Sperm.
- 9 Ovum.
- 10 Progeny.
- 11 Reciprocal cross.

36 Mitosis

Order is: first text 'The longest phase...' and illustration (b), second text 'Spindles form...' and illustration (c), third text 'When all chromatids...' and illustration (d), and last text 'The final phase...' and illustration (a), then 'Summary'

37 Meiosis

- 1 Both involve cells; the cells divide; both have a prophase, a metaphase, an anaphase and a telophase, etc.
- 2 Meiosis is more complex, it has two divisions, and gives genetically different daughter cells.
- 3 Prophase, metaphase, anaphase and telophase.
- 4 Mitosis.
- 5 To help geneticists.
- 6 Sister chromatids/homologues.
- 7 False.
- 8 False.
- 9 True.
- 10 True.
- 11 False.
- 12 True.
- 13 False.
- 14 False.

38 DNA and Nucleotides

- 1 Carbon (C), nitrogen (N), oxygen (O) and hydrogen (H).
- 2 Purines have two rings, pyrimidines have only one.
- 3 Teacher to check.
- 4 Teacher to check.
- 5 Teacher to check. ACTG should line up with TGAC.
- 6 Adenine and thymine link together 1 : 1, as do cytosine and guanine.
- 7 1 000 000 000, a thousand million or a (US) billion.
- 8 No, it can produce a favourable characteristic (but rarely).
- 9 Weak ('loosely linked').
- 10 A sugar.
- 11 Oxygen has been removed.

39 Inheritance

- Cross 1** Parents: TT and tt.
F₁ genotypes: all Tt.
F₁ phenotypes: all straight-tailed.
- Cross 2** Parents: Tt and Tt.
F₂ genotypes: TT, Tt, Tt, tt.
F₂ phenotypes: three straight-tailed, one curly-tailed.
- Cross 3** Parents: ee and EE.
F₁ genotypes: all Ee.
F₁ phenotypes: all round ears.
- Cross 4** Parents: Ee and Ee.
F₂ genotypes: EE, Ee, Ee, ee.
F₂ phenotypes: three round-eared, one oval-eared.
- Cross 5** Parents: WW and ww.
F₁ genotypes: all Ww.
F₁ phenotypes: all long-whiskered.
- Cross 6** Parents: cc and CC.
F₁ genotypes: all Cc.
F₁ phenotypes: all smooth-coated.
- Cross 7** Parents: Cc and Cc.
F₂ genotypes: CC, Cc, Cc, cc.
F₂ phenotypes: three smooth-coated, one rough-coated.

40 Genotypes, Phenotypes and Crosses

- 1 When dominant alleles and recessive alleles are both present, only one dominant allele is needed for that trait to show.
- 2 F₁ is the first filial generation (children), F₂ is the second filial generation (grandchildren).
- 3 Genotype is the (invisible) genetic constitution, for example homozygous tall plants are TT, etc. Phenotype is the (visible) physical characteristics, for example tall, or dwarf.
- 4 F₂ genotypes are all Aa and phenotypes are all brown fur. F₂ genotypes are AA, Aa, Aa and aa. Phenotypes are three brown fur, and one white fur.
- 5 Environment, such as food, etc.
- 6 Where only one characteristic is involved.
- 7 Mendel.
- 8 Cystic fibrosis, sickle-cell anaemia, haemophilia, colour blindness, Huntington's chorea, etc.
- 9 There are two factors for each characteristic an individual inherits.
- 10 It takes many (40 to 50) years for symptoms to appear.
- 11 If the parent is a heterozygous carrier of the condition, a child need not inherit the gene for the disease from the parent.

41 Evolution: survival of the fittest

- 1 A longer tail could slow movement, or make it easier for a predator to catch the animal.
- 2 If features become too elaborate they could become counter-productive and this means the individual is less viable.
- 3 Behaviour, such as a dance, a courtship ritual, a call, nest-making, etc.



Answers

- 4 Dress (fashion), behaviour, etc. Could mention high heels, body-piercing, tattoos, etc.
- 5 In fighting with predators, or other male deer.

42 Sex and Genes

- 1 Two XX and two XY
- 2 1 : 1
- 3 23
- 4 In mature males: facial hair, body hair, deep voice, etc.
- 5 One that is 'overpowered' or 'hidden' when matched with a dominant gene (allele).
- 6 X^cX , X^cY , XX, XY
- 7 X^cX is female, with normal colour vision, X^cY is male and colour blind, XX is female, with normal colour vision, XY is male, with normal colour vision.
- 8 Someone who carries the recessive gene but doesn't show symptoms of disease.
- 9 Two X^cX^R , two X^cY .
- 10 Two X^cX^R females, with red eyes, two X^cY males, with white eyes, that is a ratio of 1 : 1.
- 11 DMD, haemophilia, etc.
- 12 They breed quickly, are easy to maintain, etc.
- 13 Fruit flies.
- 14 Sex linked diseases are attached to a particular (usually X) chromosome and are passed on by one parent only. Recessive genes can be passed on by either male or female parent.
- 15 Blood doesn't clot/coagulate, leading to (internal) bleeding.
- 16 True.

43 Karyotypes

Teacher to check that the sizes and banding patterns match. The largest pair is to be numbered 1, and the pairs are to be in order of size down to the smallest.

44 *Drosophila melanogaster*: the geneticist's favourite pet

Answers will vary due to the random nature of shuffling cards. Teacher to check.

45 GMOs: why all the fuss?

- 1 Advantages: 'better' traits than conventional crops or animals, characteristics similar, predictable outcomes, ready market exists for the product, etc.
- 2 Disadvantages: too similar, can be wiped out by disease, unknown problems could occur, loss of 'wild type' characteristics, ownership of organisms by firms rather than the public.
- 3 wool, meat, size, fine fibre.
- 4 Teacher to check.

46 Sex Chromosomes and the Human Body Plan

- 1 XX
- 2 XY
- 3 Male (the father)
- 4 Carries both the male Y chromosome and the female X chromosome.
- 5 Fertilised ovum.
- 6 Must be able to produce a reproductive tract capable of delivering live offspring.
- 7 chromosomes
- 8 one
- 9 male
- 10 duct
- 11 female
- 12 hormones
- 13 Wolffian

- 14 a Developing male.
b Early stages of development.
c Developing female.

47 Chromosomes and Gene Mapping

- 1 Homologous pair.
- 2 Allele
- 3 Allele pair.
- 4 Gene locus.
- 5 Homologous pair
- 6 Allele pair
- 7 Locus
- 8 Allele
- 9 G
- 10 A
- 11 C
- 12 F
- 13 Eye colour.
- 14 Hair colour.
- 15 Hair development.
- 16 Wing development.

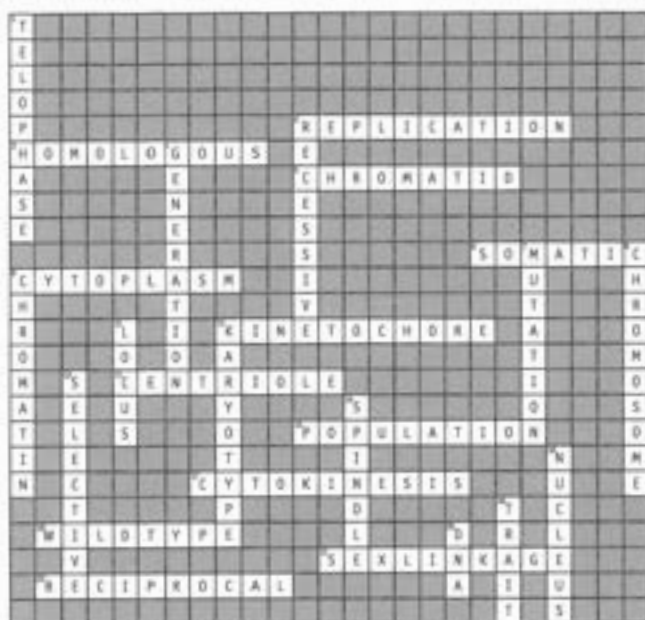
48 Hot Issues in Genetics: a research project

- 1 Something that might happen, the consequences, or ramifications, etc.
- 2 Principles, standards, morals, code of conduct, etc.
- 3 Availability of facilities to do the transplant, suitability of recipient, other needy recipients, compatibility of donor organ, etc.
- 4 Greater knowledge helps us understand how the body works, helps identify possible health problems in an individual, helps parents make decisions as whether or not to have children, etc.
- 5 Discriminate against individuals, for example in health care, insurance, etc; could create prejudices, etc.
- 6 Improves quality of life, cures disease, could prevent transmission of undesirable traits to next generation, etc.
- 7 Still experimental, could cause unforeseen problems, could raise expectations unreasonably, cost, etc.
- 8 Choose from cystic fibrosis, muscular dystrophy, etc.
- 9 Parents get what they want, possibly healthier babies, etc.
- 10 Imbalance in sex of population, still experimental, could cause unforeseen problems, could raise expectations unreasonably, cost, etc.
- 11 In-vitro fertilisation (IVF) involves a sperm and an ovum being taken from parents and united outside the body, and then being raised in an artificial environment, usually a petri dish. When a zygote is viable, it is re-implanted in the mother's womb to continue its development.
- 12 Because one or other partner is infertile or the parents are having difficulty conceiving, etc.
- 13 It is not 'natural', parents are not always suitable, medical resources could be better used elsewhere, etc.
- 14 A valuable individual could be replaced, we can grow (compatible) replacement organs, etc.
- 15 The loss of genetic variation, it is expensive, medical resources could be better used elsewhere, etc.
- 16 An unspecialised cell that gives rise to differentiated cells. It has not committed to what it will become yet.
- 17 Replacement body parts can be 'grown', faulty organs can be fixed, etc.
- 18 As in many areas of medical research concerns over ethics, cost, chances of success, etc. The main issue is that currently the research requires a dead/aborted foetus. This raises ethical, religious concerns and concern that people may sell their foetus for research.



Answers

49 Another Genetics Crossword



50 The Hardy-Weinberg Law

- Mention of fair test, statistically significant numbers, no outside influence, etc.
- Genotypes are AA, Aa, Aa and aa.
- 3 out of 4 contain A and 3 out of 4 contain a.
- It can have many generations in a very short time.
- 89.
- $p = 29 \text{ AA} + 0.5 \times 44 \text{ Aa} = 29 + 22 = 51$,
Percentage = $\frac{51}{89} = 57\%$
- Assume q must make up the remainder,
hence $(100 - 57)\% = 43\%$
- $q = 0.5 \times 44 \text{ Aa} + 16 \text{ aa} = 22 + 16 = 38$,
Percentage = $\frac{38}{89} = 43\%$
- 45.
- $p = 20 \text{ AA} + 0.5 \times 18 = 20 + 9 = 29$,
Percentage = $\frac{29}{45} = 64\%$
- $q = 0.5 \times 18 \text{ Aa} + 7 \text{ aa} = 9 + 7 = 16$,
Percentage = $\frac{16}{45} = 36\%$
- 29.
- 44.
- 16.
- 20.
- 18.
- 7.
- Both alleles are dominant.
- The two alleles are different, one dominant and one recessive.
- Both alleles are recessive.
- True.

51 Gene Expression: a look at cats

- How well it is fed will affect size, etc.
- Albinism.
- No, chromosomes are inactivated at random.
- Fur colour is sex-linked (determined by genes on the X chromosome).
- Must be female, (yellow and black gene each found on different X chromosomes).
- Yes (black on one X chromosome and white activation can be linked to either yellow or black in the presence of one X only).
- To produce appealing breeds.

52 Coming to a Conclusion

- False, proteins are made up of amino acids and hence nucleotides.

- True. Generally lethal genes must be double dose (homozygous) or dominant, but sex-linked cases do occur.
- False, it can cause less fit organisms, but it can also give favourable traits.
- True, bringing together/preserving rare alleles instead of mating 'normal' dogs.
- False, only links between adenine/thymine and cytosine/guanine exist.
- True, small molecule links with a large molecule on the other strand, otherwise the DNA double helix would be unbalanced.

53 Looking for Clues

- | | |
|---------------|-------------------------------------|
| 1 Chromatid | 15 Locus |
| 2 Heredity | 16 Hybrids |
| 3 RNA | 17 Genotype |
| 4 Offspring | 18 F_1 , heterozygous, monohybrid |
| 5 Mitosis | 19 Filial |
| 6 O | 20 Heredity |
| 7 Sexual | 21 Meiosis |
| 8 Ovulation | 22 Phenotype |
| 9 Meiosis | 23 Chromosome |
| 10 Evolution | 24 Nuclear |
| 11 CHROMOSOME | 25 Hybrid |
| 12 Somatic | 26 Somatic |
| 13 Purebred | 27 Mitosis |
| 14 Germ | 28 Allele |

54 Parents and Offspring

- Parent 1 (BB)
- Parent 2 (bb)
- Heterozygous, Bb
- No, all Bb.
- BB, BB, Bb, Bb. All brown.
- Bb, Bb (brown) bb, bb (blue).
- Only one recessive allele in the first cross, therefore hidden. Two recessive alleles in second cross, therefore could be expressed.

55 Blood Groups

- Alternative form of a gene.
- Four.
- $I^A I^B = AB$, $I^B I^B = B$, $I^A I^O = A$, $I^B I^O = B$.
- $I^A I^A = A$, $I^A I^A = A$, $I^A I^B = AB$, $I^A I^O = A$.
- Karl Landsteiner.
- Paternity cases, scene of crime, etc.

56 Sutton, Boveri and Chromosomes

- Structure and function of cells.
- Making of gametes (sex cells).
- Greek words meaning 'coloured body'.
- They (independently) came up with the currently held theory.
- Genes.
- See the points in text provided. Teacher to check.
- Drosophila melanogaster*, the fruit fly.

57 Who (or What) am I?

- | | |
|-------------------|----------------|
| 1 Gregor Mendel. | 11 Chromosome. |
| 2 W. S. Sutton. | 12 Fruit fly. |
| 3 T. H. Boveri. | 13 Pea. |
| 4 K. Landsteiner. | 14 Sperm. |
| 5 W. Johannsen. | 15 DNA. |
| 6 R. C. Punnett. | 16 Gene. |
| 7 J. Watson. | 17 Allele. |
| 8 A. R. Wallace. | 18 Nucleus. |
| 9 C. Darwin. | 19 Nucleotide. |
| 10 W. Bateson. | 20 RNA. |



Answers

58 More Genetics Terms

- | | | |
|-----|------|------|
| 1 c | 10 v | 19 l |
| 2 f | 11 a | 20 u |
| 3 z | 12 b | 21 m |
| 4 x | 13 d | 22 r |
| 5 o | 14 e | 23 n |
| 6 t | 15 g | 24 p |
| 7 w | 16 h | 25 q |
| 8 y | 17 i | 26 s |
| 9 j | 18 k | |

59 Genetics Assignment

Answers will vary for Questions 1 to 5, but should include the following.

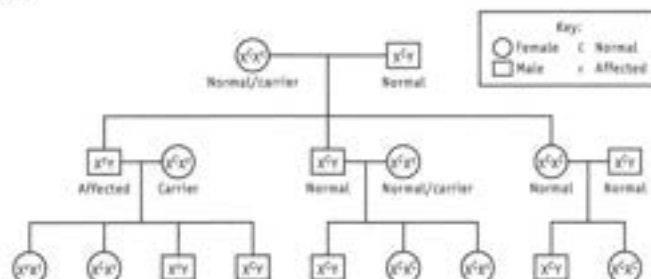
- Worked together in the UK in the 1950s on the structure of DNA.
- Worked independently on chromosomes, Sutton in the USA in the early 1900s and Boveri in Germany.
- Worked independently on evolution/natural selection. Darwin in Australasia and South America, was published 1858 (after a long delay). Wallace in South America and Malay Peninsula, also published 1858.
- Worked independently on population genetics. Around 1950, Hardy in the UK, Weinberg in Germany.
- Worked together on genetics of microorganisms (mutations in mould, etc), in the USA in the 1940s and 1950s.
- To do with chromosomes that are not sex chromosomes.
- A molecular chain consisting of amino acids.
- When similar organisms in different environments gradually change characteristics.
- The study of past times using fossils.
- Breeding individuals with specific traits to emphasise a characteristic.

60 Fertility: seedless watermelons and the mule

- Edible reproductive part of a seed plant (usually has seeds surrounded by a pulp).
- Having the 'normal' chromosome number, $2n$.
- The ability to reproduce.
- Group of similar animals able to breed with each other and produce fertile offspring.
- Hybrids, formed by breeding two different species.
- Used by humans.
- Female horse.
- Male horse.
- Unable to reproduce.
- Young horses.
- External interference, an outside influence else being involved.
- The marked energy and capacity for growth often shown by crossbred organisms.
- To combine the best properties from both parent animals.
- Donkey.
- Grapes.
- Chromosomes are too different to pair up and to produce viable gametes.

61 Genetics Test

- B
- B
- D
- B
- C
- D
- A
- C
- B
- D
- C
- A
- D
- C
- B
- A
- True.
- True.
- False.
- False.
- True.
- False.
-



- They are all female.
- It is carried on the X chromosome (sex-linked) and males are affected by a single dose.
- Contracting a disease involves catching it through a microorganism such as a virus or bacteria. Inheriting a disease is through genes passed on from a parent. External source versus an internal source.
- To ensure diversity and to ensure survival of at least some of the species if conditions change.
- For: Can advance our knowledge of the human body, could provide replacement body parts. Against: ethical considerations, could reduce the size of the gene pool, etc.

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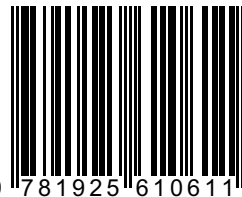
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